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S.I.: RECENT DEVELOPMENTS IN FINANCIAL MODELING AND RISK MANAGEMENT

Forecasting commodity futures returns with stepwise regressions: Do commodity-specific factors help?

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Abstract
The aim of this paper is to assess whether three well-known commodity-specific variables (basis, hedging pressure, and momentum) may improve the predictive power for commodity futures returns of models otherwise based on macroeconomic factors. We compute recursive, out-of-sample forecasts for the monthly returns of fifteen commodity futures, when estimation is based on a stepwise model selection approach under a probability-weighted regime-switching regression that identifies different volatility regimes. We systematically compare these forecasts with those produced by a simple AR(1) model that we use as a benchmark and we find that the inclusion of commodity-specific factors does not improve the forecasting power. We perform a back-testing exercise of a mean–variance investment strategy that exploits any predictability of the conditional risk premium of commodities, stocks, and bond returns, also consider transaction costs caused by portfolio rebalancing. The risk-adjusted performance of this strategy does not allow us to conclude that any forecasting approach outperforms the others. However, there is evidence that investment strategies based on commodity-specific predictors outperform the remaining strategies in the high-volatility state.

Keywords Stepwise regression · Commodity returns · Predictability · Portfolio back-testing

1 Introduction

Recently, in the academic literature a debate has raged on the role played and the alleged distortions induced by speculative investors in commodity markets. For instance, Tang and Xiong (2012) have reported a surge from \$15 to \$200 billion of capital flows in commodity futures markets from institutional investors between 2003 and 2008. Irwin and Sanders (2011), Tang and Xiong (2012), and Hamilton and Wu (2015) have traced

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the start of the process of financialization of commodity markets back to 2004. Following this increasing interest in commodities as an asset class, a literature has emerged that investigates the risk sources driving commodity futures returns. In this paper we formally test—both using standard statistical criteria and resorting to economically grounded loss functions—whether variables that describe the dynamics and state of (dis-)equilibrium in the very commodity markets are needed to model and forecast commodity returns or this information is already captured by traditional macroeconomic variables (as conjectured, for instance, by Bessembinder and Chan 1992).

Our paper builds on two different strands in the literature on commodities. The first strand argues that the expected return of any given commodity futures is driven by factors that are specific to each market. This group of papers is based on two main classical theories, which posit that the level of inventories and the relative positions of short versus long hedgers in the futures markets (hedging pressure) are the key determinants of futures returns. First, the theory of storage (Kaldor 1939; Brennan 1958) hypothesizes that producers and inventory holders receive implicit benefits from inventories deriving from the possibility to manage any temporary shortages (the "convenience yield"). However, due to the presence of costs of carrying such inventories, these benefits decline as inventories increase. Given that the level of inventories plays a role in determining the relative movements of spot and futures prices, the convenience yield turns out to be related to the *basis*, i.e., the difference between spot and futures prices. Second, the theory of normal backwardation in Keynes (1930) and Hicks (1939) is based on the intuition that, to induce risk-averse speculators to take long positions, current futures prices must be set at a discount versus expected future spot prices at maturity, i.e., in ordinary times, the market would be in backwardation. The size of the discount is the future risk premium and it depends on the interplay between hedgers and speculators. Many papers have provided empirical evidence on the forecasting power of commodity-specific factors related to these classical theories (basis and hedging pressure, besides momentum that may be associated to speculative herding behavior). For example, Gorton et al. (2012), exploiting the Theory of Storage, identify a linkage between the basis and individual commodity futures returns. Building on the theory of hedging pressure, Stoll (1979), Hirshleifer (1988) and de Roon et al. (2000) found that hedging pressure affects individual commodity futures premiums.

A second strand of literature has tested the relationship between macroeconomic variables and commodities. These papers are mainly based on the assumption that storage costs and convenience yields are expected to be influenced by the general state of the economy through short-term mismatches between the demand and supply of commodities (see e.g., Bessembinder and Chan 1992). However, only a small portion of the literature studies the predictive power of macroeconomic variables that have been proven to carry a relation with commodity futures returns. For instance, Gargano and Timmermann (2014) examine the predictability of commodity returns from some macroeconomic variables and find that bond spreads, the growth rate of money supply, and industrial production are better predictors for raw industrials and metals index returns, than for foods and textiles commodities. They also observe that the predictability of commodity futures returns based on inflation, industrial production, and money supply is stronger during economic recessions than during expansions. Giampietro et al. (2018) test whether flexible specifications of pricing kernels that jointly price the cross-section of commodities, equities, and government bonds reflect standard macroeconomic variables or, alternatively, the stochastic discount factor should include commodity-specific variables as well; they find evidence that commodity market information would be required.

With these two competing strands of literatures in mind, in this paper we investigate whether models extended to include three commonly used aggregate commodity-specific factors (i.e., basis, hedging pressure, and momentum) display stronger predictive power than models based on the lagged values of 128 macroeconomic-based factors, as in Ludvigson and Ng (2009) and Welch and Goyal (2008). Our question is of considerable importance to both practitioners (arguably, investors, including both hedgers and speculators) and academics: a rejection of the null hypothesis that commodity returns can be forecast by a restricted model that only contains macroeconomic factors would represent robust evidence in favor of the segmentation of commodities from the rest of the asset classes. Indeed, if accurate and economically valuable predictions could be derived only from an extended model that includes the information in commodity-specific variables, this would confirm that commodities constitute a source of diversification to a portfolio manager, as they would offer exposures to factors that are specific to this asset class.

We conduct our analysis on 15 commodity futures series, i.e., Brent Crude Oil, Gasoline, Corn, Soybeans, Wheat, Coffee, Cocoa, Sugar, Cotton, Gold, Silver, Platinum, Orange Juice, Lumber, and Live Cattle. We examine the sample period January 1989–December 2012. As You and Daigler (2013), we focus on individual futures contracts. To assess out-of-sample (henceforth, OOS) predictability, we split our data and use information for the period January 1989–December 2003 for in-sample estimation of the models, and data for the period January 2004–December 2012 to test the recursive OOS forecasting performance of a range of alternative models. In addition, we consider different regimes of market volatility (proxied by either the VIX or by the variance of the unexplained returns from the very predictive models) and investigate whether the predictive power of different models may depend on such regimes. To test the conjecture of a strong dependence, we split both the in-sample and OOS windows in high- and low-volatility periods (differently from Jensen et al. 2000, who have instead emphasized monetary policy regimes). To reduce the number of macroeconomic variables that need to be considered, we adopt a principal components analysis, as suggested by Stock and Watson (2006): we therefore identify ten linearly uncorrelated variables, which effectively summarize 60% of the variance of the initial information set. In addition, to decide whether all these ten variables that we have constructed should be actually included in predictive regression for the returns of each of the commodities, we exploit the flexibility offered by forward and backward stepwise regressions, a tool that allows some (or all) of the variables to be chosen automatically, using various statistical criteria (see, e.g., Sharma and Yu 2015). This approach allows us to obtain the “best” subset of variables for each commodity series, thus avoiding statistically irrelevant predictors, which would just add noise to the forecasting exercises, uselessly reducing the degrees of freedom. Because our main goal is to assess whether commodity-specific variables improve the OOS predictive ability of such flexible, macro-based models, we also estimate predictive regressions where these factors are added to the models previously specified. We compare OOS forecasts from a model based on the principal components of macroeconomic variables only versus forecasts produced using also the commodity-specific factors. To evaluate the forecasting accuracy of the alternative models, we adopt two measures: the mean absolute error (MAE) and the root mean square forecast error (RMSFE). Both criteria lead to the conclusion that neither the models that include only macroeconomic variables nor the extended models that contain also commodity-specific factors yield better OOS performances than a simple, naïve first-order autoregressive benchmark.

Finally, we turn to the assessment of the economic value of alternative predictions (i.e., those based on macroeconomic variables only versus those that include commodity-specific factor as well) using a mean–variance framework (similar to Jensen et al. 2000; Erb and Harvey 2006; Fuertes et al. 2010), in which the asset menu includes both commodities and traditional assets (equities and bonds). Although such a OOS portfolio back-testing may seem not justified by the in-sample and statistical OOS results, a recent literature in portfolio management (see, e.g., Dal Pra et al. 2018) has shown that—because the typical loss functions employed are deeply different—it is possible for statistical back-testing to reveal a distorted, downward-biased picture of the amount of economic value that instead a OOS portfolio back-test may disclose. In fact, our results suggest that portfolios resulting from the joint use of macroeconomic and commodity-specific factors perform better (in the case of models built by forward selection) than models driven by macroeconomic variables only also when transaction costs are considered. However, this result appears to be reversed when regressions are built using backward selection stepwise methods. Interestingly, the asset allocations are dominated by long exposures to bonds and the residual is structured as a long/short strategy with an almost perfect zero net exposure to commodities and equities. This empirical result is in line with a recent literature that has showed that, even going beyond standard mean–variance analysis, it may be difficult to obtain evidence in favour of multi-asset portfolios including long positions in individual commodities (see, e.g., Henriksen et al. 2019; Lean et al. 2018). Finally, it is interesting to notice that when these analyses are performed separately with reference to low- versus high-volatility regimes, we find that in the high-volatility regime models that include commodity-specific factors imply more accurate forecasts, higher realized Sharpe ratios, and higher mean–variance realized utility values than the low-volatility regime portfolios. These results appear novel: You and Daigler (2013) have examined the diversification benefits of using individual futures contracts in a Markowitz framework and investigated the differences between ex-ante, in-sample results and ex-post, realized performances. However, our focus is distinctively devoted to the differential predictability of commodity-specific versus variables that capture the state of the business cycle.

The rest of the paper is organized as follows: Sect. 2 describes the methodology. Section 3 contains a description of the data. Section 4 reports the results of the estimation of the models and their OOS performances. Section 5 reports the back-testing results from the mean–variance exercises. Section 6 concludes.

2 Research design

2.1 Definition of volatility regimes

It is nowadays well-known in the literature that different regimes of market volatility should be featured in factor models, based on the evidence that financial markets may change dramatically (see, e.g., Rapach and Zhou 2013). Portfolio managers, risk arbitrageurs, and corporate treasurers closely monitor volatility trends, because changes in prices could have a major impact on their investment and risk management decisions. We adopt the VIX index provided by the Chicago Board Options Exchange (CBOE) from January 1989 to December 2012 as a proxy of market volatility. Given that the VIX index is characterized by the presence of different states (or regimes), we introduce a regime switching framework in order to disentangle the different levels of market volatility. In particular, we

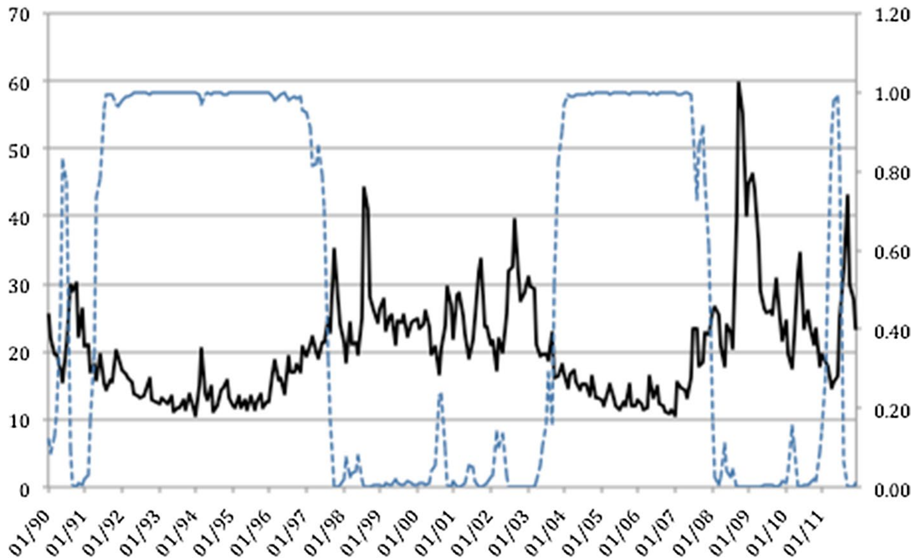

Fig. 1 VIX index series and estimated filtered probability of being in the low-volatility state

adopt the following two-state regime switching model to feature the existence of a high-volatility and a low-volatility state[1]:

$$VIX_t = c_{s_t} + \phi VIX_{t-1} + \varepsilon_t, \tag{1}$$

with $\varepsilon_t \sim \mathrm{N}\left(0, \sigma^2\right)$. The variable s_t is a Markov state variable, so that the mean, c, of the VIX process takes different values (say, c_{high} and c_{low}) in the two different regimes and the transition between the two regimes is governed by a Markov process:

$$P = \begin{bmatrix} P\left(s_t = 0|s_{t-1} = 0\right) & P\left(s_t = 1|s_{t-1} = 0\right) \\ P\left(s_t = 0|s_{t-1} = 1\right) & P\left(s_t = 1|s_{t-1} = 1\right) \end{bmatrix} = \begin{bmatrix} p_{00} & p_{10} \\ p_{01} & p_{11} \end{bmatrix}, \tag{2}$$

where p_{ij} $(i, j = 0,1)$ denotes the transition probability of $s_t = j$, given $s_{t-1} = i$ and the transition probabilities satisfy $p_{i0} + p_{i1} = 1$. The transition matrix governing the behaviour of the state variable, contains only two parameters p_{00} and p_{11}. Hence, we assume that the variable s_t is not directly observable, but that it can be inferred from the past behaviour of VIX_t.

Figure 1 plots the estimated filtered probabilities that the VIX was in a low-volatility regime (clearly the plot for high-volatility state is just specular). From an inspection of the figure, it seems that the entire sample period could be divided into four different sub-periods. In particular, January 1989-December 1997 and January 2004-December 2007 can be considered as low-volatility periods, while January 1998–December 2003 and January 2008-December 2012, as high-volatility periods.

[1] Jensen et al. (2000) provide evidence on the role of commodity futures in mean–variance portfolios. They find that in periods of restrictive monetary policy, commodity futures carry an important weight and yield a considerable performance enhancement. However, since their paper, it has become common to classify financial market regimes on the basis of the level of volatility.

2.2 Forecasting with many macroeconomic predictors: principal components

Stock and Watson (2006) surveyed the literature concerning methods to forecast economic time series variables using many predictors. Among all the methodological solutions analysed, the authors showed that forecasts based on principal components of a large number of predictors are first-order asymptotically efficient given that both the number of predictors and the number of observations are very large, and this is particularly true when we consider a large set of macroeconomic variables (Stock and Watson 2002). The core idea of principal component analysis (PCA) is to identify a limited number of linearly uncorrelated variables, the so-called principal components (PCs), which summarize the largest part of the variation of a sample set of potentially correlated variables. The PCA approach is based on a well-known result, the *singular value decomposition theorem*, that states that any symmetric matrix, and therefore also a variance–covariance matrix Σ, can be decomposed through the symmetric eigenvalue decomposition as

$$\Sigma = U\Lambda U', \tag{3}$$

where U is a square $n \times n$ matrix whose ith column is the eigenvector u_i of Σ and Λ is a diagonal matrix whose diagonal elements are the corresponding eigenvalues, $\Lambda_{ii} = \lambda_i$. Notably, U is an orthogonal matrix, so that $UU' = U'U = I_n$. After obtaining the decomposition of the variance–covariance matrix of our n starting variables, we compute the principal components:

$$PC_k = \sum_{i=1}^{n} u_{ki} y_i, \tag{4}$$

where $y_1, y_2, \ldots, y_n$ represent the vectors of the original variables (in our case the 128 macro-based factors that we will describe in details in Sect. 3) and u_k is the kth eigenvector of Σ and $u_{k1}, u_{k2}, \ldots, u_{kn}$ are its n elements.

Using principal component analysis to summarize a large set of macroeconomic factors, we estimate the following model for the returns of each commodity j, $R_{t+1,j}$:

$$R_{t+1,j} = \alpha_j + \sum_{i=1}^{N} \beta_{ij} PC_{t,ij} + D_{t,j}\left(\sum_{k=1}^{K} \gamma_{kj} C_{t,kj}\right) + \varepsilon_{t,j}, \tag{5}$$

where α_j is the model's intercept, β_{ij} is the factor loading of the ith principal component for commodity j, γ_{kj} represents the regression coefficients of the kth commodity-specific factor $C_{t,kj}$, and $D_{t,j}$ is a dummy variable, which is equal to 1 when the model also contains commodity-specific factors, and 0 otherwise. $\varepsilon_{t,j}$ represents the error term, assumed to be white noise and such that $Corr(\varepsilon_j, \varepsilon_i) = 0$, for all pairs i and j. Figure 2 reports the graph of the cumulative variance explained by principal components. To reduce the number of parameters to be estimated, we decided to include only the first ten components resulting from the PCA performed, which explain 60% of the total variance of the initial informational set (the dotted line in the Fig. 2 represents the tenth principal component). In addition, Table 1 provides a representation of the scale of the factor loadings. They range from -1 to 1; loadings close to -1 or 1 indicate that the factor is strongly correlated with the variable, while loadings close to zero indicate that the link is weak.

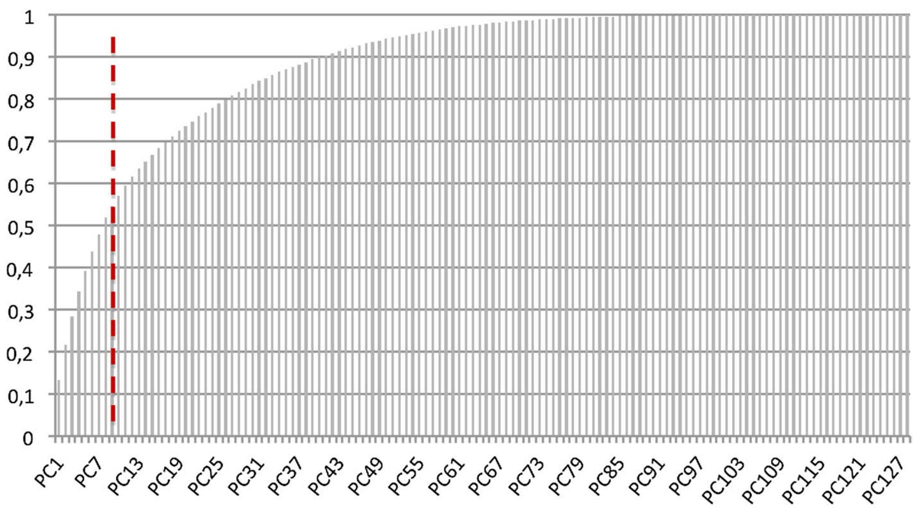

Fig. 2 Cumulative proportion of explained variance by principal components

2.3 Variables selection method: stepwise regressions

Starting from the set of principal components computed previously, we apply a variable selection methodology to decide the variables to be included as regressors to estimate futures returns of each commodity in the sample. To this purpose we rely on stepwise regression, an automatic variable selection procedure, which chooses from a set of candidates the explanatory variables that are, jointly, the most relevant. Different stepwise regression procedures are available, but we decided to use the unidirectional forward and backward methods. Forward selection starts with no variables in the model, testing the inclusion of each variable with a chosen model-fit criterion, adding the variable (if any) whose inclusion gives the most statistically significant improvement of the fit, and repeating this process until none of the remaining variables improves the model to a statistically significant extent. Backward elimination starts with all candidate variables, testing the deletion of each variable using a chosen model-fit criterion, deleting the variable (if any) whose exclusion gives the least statistically significant deterioration of the model fit, and repeating this process until no further variables can be deleted without a statistically significant loss of fit.

Stepwise regression procedures admit different selection criteria for variables to be included or excluded from the models; for instance, it may rely on a sequence of F-tests or on an information criterion, which is a measure that trades-off in-sample fit with parsimony of the model, such as the Akaike information criterion (henceforth, AIC). In line with the recent literature, we adopt the AIC as a selection criterion, $AIC = 2k - 2\ln(\hat{L})$, where $\hat{L}$ is the maximum value of the likelihood function of the model and k is the number of parameters to be estimated. Given a set of models, the preferred one is that with the smallest AIC value.[2]

[2] Yamashita et al. (2007) compare the stepwise AIC selection method with other stepwise methods for variable selection and show that this practical criterion leads to the same results as partial F tests.

Table 1 Principal components factor loadings

Variable	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10
Production Index	-0.04	0.07	-0.03	0.01	-0.04	0.05	-0.02	0.14	-0.01	0.01
Production Index less transfers	-0.11	0.10	-0.02	0.07	-0.02	0.12	-0.08	0.14	0.08	-0.07
Real Consumption	-0.01	0.10	-0.10	0.05	0.02	-0.01	0.15	0.05	-0.12	0.13
Manufacturin & Trade sales	-0.06	0.09	-0.12	0.04	0.14	-0.08	0.09	-0.03	-0.11	0.22
Retail sales	0.01	0.09	-0.11	0.04	0.06	-0.06	0.15	0.10	-0.18	0.12
Internal Production : total	-0.19	0.05	-0.04	-0.03	0.20	0.03	-0.01	-0.02	0.01	-0.04
Internal Production : products	-0.17	0.04	-0.05	-0.02	0.22	0.06	-0.01	-0.05	0.05	-0.10
Internal Production : final prod	-0.15	0.04	-0.05	-0.02	0.23	0.07	0.00	-0.08	0.04	-0.11
Internal Production : consumer goods	-0.12	0.05	-0.05	-0.04	0.23	0.04	0.00	-0.13	0.06	-0.11
Internal Production : durable consumer goods	-0.12	0.07	-0.05	-0.06	0.22	0.05	-0.04	-0.11	0.00	-0.06
Internal Production : cons nondble	-0.06	0.00	-0.02	0.00	0.11	0.01	0.03	-0.08	0.09	-0.10
Internal Production : bus eqpt	-0.17	0.01	-0.05	0.01	0.15	0.09	-0.01	0.01	0.00	-0.07
Internal Production : materials	-0.18	0.05	-0.03	-0.03	0.13	-0.01	0.00	0.02	-0.04	0.06
Internal Production : durable goods materials	-0.18	0.06	-0.01	-0.01	0.14	0.01	-0.02	0.03	-0.01	0.01
Internal Production : nondurable goods materials	-0.09	0.00	-0.02	-0.07	0.05	-0.11	0.00	0.09	-0.08	0.01
Internal Production : manufacturing	-0.19	0.05	-0.04	-0.03	0.20	0.02	-0.02	0.02	0.00	-0.04
Internal Production : residential utilities	0.03	0.17	0.20	-0.02	0.05	0.04	0.16	-0.02	0.03	-0.03
Internal Production : fuels	0.00	0.00	-0.03	-0.05	0.09	-0.02	0.09	0.07	-0.03	0.04
NAPM production index	-0.19	0.04	0.08	0.00	-0.07	-0.11	-0.04	0.00	0.04	0.00
Capacity utilization	-0.12	-0.14	-0.10	0.12	-0.05	0.14	-0.05	0.06	-0.06	0.09
Index of Help-wanted advertising	-0.10	0.03	0.02	-0.01	0.03	0.02	-0.10	0.12	0.21	-0.06
Ratio of Help-Wanted Ads/No. Unemployed (AC)	-0.14	0.03	0.02	0.00	0.02	0.04	-0.10	0.10	0.19	0.00
Civilian Labor Force: Employed, Total	-0.09	-0.01	0.05	0.01	-0.05	0.03	-0.06	0.15	0.04	0.08
Civilian Labor Force: Employed, Nonagric.Industries	-0.08	0.00	0.05	0.01	-0.05	0.04	-0.05	0.15	0.04	0.08
Unemployment Rate: All Workers, 16 Years & Over	0.12	-0.02	-0.01	-0.02	0.01	-0.04	0.03	-0.01	-0.07	-0.10
Unemployment By Duration: Average Duration In Weeks	0.05	0.04	-0.02	-0.08	-0.08	-0.09	-0.05	0.05	0.01	-0.09
Unemploy By Duration: Persons Unempl Less Than 5 Wks	0.03	-0.03	-0.02	0.05	0.09	-0.03	0.09	0.00	-0.03	0.02
Unemploy By Duration: Persons Unempl 5 To 14 Wks	0.06	-0.01	0.00	-0.01	-0.03	0.02	-0.02	-0.03	-0.04	-0.09
Unemploy By Duration: Persons Unempl 15 To 26 Wks	0.08	0.01	-0.01	-0.06	-0.08	0.01	-0.05	0.01	-0.03	-0.09
Initial Claims for Unemployement Insurance	0.07	-0.07	0.07	0.04	-0.04	0.03	-0.04	-0.02	-0.03	-0.04
Employees On Nonfarm Payrolls - Goods-Producing	-0.19	-0.01	-0.01	0.09	0.01	0.09	-0.02	0.09	-0.06	0.09
Employees On Nonfarm Payrolls - Mining	-0.01	0.01	0.00	0.01	0.03	0.13	0.04	0.07	-0.03	-0.03
Employees On Nonfarm Payrolls - Manufacturing	-0.19	-0.07	-0.02	0.10	0.02	0.07	-0.06	-0.01	-0.06	0.05
Employees On Nonfarm Payrolls - Durable Goods	-0.19	-0.03	0.00	0.10	0.03	0.09	-0.05	-0.04	-0.06	0.06
Employees On Nonfarm Payrolls - Nondurable Goods	-0.14	-0.14	-0.07	0.06	-0.01	-0.03	-0.08	0.08	-0.03	-0.01
Employees On Nonfarm Payrolls - Financial Activities	-0.01	0.06	0.02	0.05	0.02	0.15	-0.01	0.03	-0.04	0.07
Employees On Nonfarm Payrolls - Government	0.01	0.02	-0.05	-0.05	-0.05	0.03	-0.02	-0.03	-0.07	-0.02
Avg Weekly Hrs of Prod - Goods-Producing	-0.05	0.04	-0.08	-0.04	0.10	-0.04	0.03	0.18	-0.02	-0.18
Avg Weekly Hrs of Prod - Mfg Overtime	-0.06	0.01	-0.04	-0.09	0.06	-0.05	0.02	0.11	-0.01	-0.17
Average Weekly Hours, Mfg.	-0.07	0.04	-0.08	-0.05	0.12	-0.04	0.02	0.13	0.02	-0.16
NAPM Employment Index	-0.18	-0.01	0.11	0.08	-0.07	0.08	0.03	0.05	0.01	0.05
Housing Starts:Nonfarm(1947-58);Total Farm&Nonfarm(1959)	0.01	0.06	-0.02	-0.01	0.00	-0.07	0.02	0.31	-0.06	-0.04
Housing Starts:Northeast	0.00	0.04	-0.02	-0.02	0.02	-0.03	0.02	0.11	-0.16	-0.02
Housing Starts:Midwest	0.02	0.04	-0.05	0.01	-0.01	-0.05	0.06	0.20	-0.08	-0.04
Housing Starts:South	0.00	0.04	-0.02	-0.03	0.02	-0.06	0.02	0.25	-0.07	0.02
Housing Starts:West	0.00	0.01	0.03	0.03	-0.03	0.01	-0.04	0.09	0.13	-0.08
Housing Authorized: Total New Priv Housing Units	0.01	0.07	-0.04	-0.02	0.03	-0.05	0.05	0.30	-0.02	0.01
Houses Authorized By Build. Permits:Northeast	-0.01	0.06	-0.05	-0.04	0.02	-0.04	0.05	0.18	-0.14	0.12
Houses Authorized By Build. Permits:Midwest	0.03	0.02	-0.01	-0.04	-0.03	-0.04	0.04	0.31	-0.07	-0.02
Houses Authorized By Build. Permits:South	0.00	0.05	-0.04	-0.02	0.05	-0.06	0.07	0.25	-0.03	-0.02
Houses Authorized By Build. Permits:West	0.00	0.04	0.00	0.05	0.01	0.01	-0.02	-0.01	0.14	0.00
Purchasing Managers' Index	-0.20	0.02	0.08	0.03	-0.09	-0.06	-0.02	0.03	0.02	0.01
Napm New Orders Index	-0.18	0.05	0.07	-0.03	-0.08	-0.11	-0.05	0.01	0.07	-0.03
Napm Vendor Deliveries Index	-0.15	0.06	0.14	0.06	-0.11	-0.10	0.06	0.01	-0.05	0.01
Napm Inventories Index	-0.12	-0.02	0.06	0.10	-0.08	0.06	0.03	0.08	0.02	0.07
Mfrs' New Orders, Consumer Goods And Materials	-0.07	0.04	-0.07	-0.03	0.10	-0.05	0.04	-0.13	0.02	0.22
Mfrs' New Orders, Durable Goods Industries	-0.06	0.04	-0.04	0.02	0.09	-0.04	0.04	-0.14	0.02	0.29
Mfrs' New Orders, Nondefense CaProduction Indextal Goods	-0.04	0.02	-0.03	0.04	0.02	-0.03	0.02	-0.09	0.07	0.24
Manufacturing And Trade Inventories	-0.11	-0.05	0.08	0.12	-0.07	0.05	-0.05	0.00	0.03	0.04
Ratio, Mfg. And Trade Inventories To Sales	0.03	-0.11	0.15	0.01	-0.15	0.09	-0.10	0.03	0.11	-0.21
Money Stock: M1	0.03	0.00	0.04	-0.15	0.03	-0.08	-0.27	0.08	0.13	0.07
Money Stock: M2	0.10	0.07	0.10	-0.06	0.09	0.21	-0.06	0.08	0.04	0.12
Money Stock: Currency held by the public	0.00	0.00	-0.01	-0.08	-0.03	-0.13	-0.12	-0.06	-0.01	0.16
Money Supply: Real M2 (AC)	0.09	0.14	0.10	-0.02	0.05	0.20	-0.13	0.06	0.02	0.11
Monetary Base, Adj For Reserve Requirement Changes	0.00	-0.01	0.06	-0.12	0.01	-0.07	-0.22	0.02	0.06	0.20
Depository Inst Reserves:Total, Adj For Reserve Req Chgs	0.01	-0.04	0.12	-0.14	0.04	0.03	-0.26	0.09	0.09	0.12
Depository Inst Reserves:Nonborrowed,Adj Res Req Chgs	0.02	-0.04	0.11	-0.14	0.03	0.03	-0.27	0.10	0.09	0.12
Commercial and Industrial Loans, All Commercial Banks	0.06	0.16	0.18	-0.02	0.04	0.11	0.16	-0.01	0.03	0.00
Change Commercial and Industrial Loans, All Commercial Banks	-0.01	-0.01	0.06	-0.03	-0.02	0.01	-0.09	0.08	-0.02	-0.10
Ratio, Consumer Installment Credit To Personal Income	0.01	0.04	0.09	0.06	-0.05	0.02	0.10	-0.06	-0.15	0.13
S&P 500	0.00	0.05	-0.17	0.16	-0.07	-0.01	0.08	0.05	0.29	0.08
S&P: indust	-0.01	0.05	-0.17	0.14	-0.07	-0.01	0.08	0.05	0.29	0.09
S&P div yield	0.00	-0.06	0.16	-0.15	0.07	0.01	-0.08	-0.05	-0.29	-0.08
S&P PE ratio	0.06	0.08	-0.19	0.08	-0.02	-0.06	0.04	0.01	0.27	0.04
DP	0.03	-0.03	-0.11	-0.12	0.02	-0.27	-0.16	-0.04	0.02	-0.07
b/m	-0.03	-0.03	0.05	-0.10	0.03	-0.01	0.02	-0.05	-0.01	-0.13
ntis	0.01	-0.02	-0.06	-0.09	0.05	0.02	-0.02	-0.04	0.01	-0.15
svar	0.00	-0.06	0.08	-0.02	0.05	-0.01	-0.05	0.02	-0.13	0.02
Fed Funds	-0.15	-0.08	0.07	0.09	-0.13	0.04	0.00	-0.02	-0.10	-0.05
Comm paper	-0.16	-0.03	0.01	-0.02	-0.17	0.08	0.02	-0.02	-0.10	-0.01
3 mo T-bill	-0.15	-0.03	-0.02	-0.01	-0.20	0.07	0.04	-0.01	-0.06	-0.02
6 mo T-bill	-0.15	-0.02	-0.04	-0.07	-0.20	0.09	0.04	-0.02	-0.05	-0.01
1 yr T-bond	-0.14	0.01	-0.07	-0.13	-0.20	0.11	0.06	-0.02	0.00	0.01
5 yr T-bond	-0.07	0.03	-0.10	-0.23	-0.16	0.11	0.09	-0.04	0.04	0.04
10 yr T-bond	-0.06	0.03	-0.08	-0.26	-0.14	0.10	0.09	-0.03	0.04	0.04
Aaa bond	-0.07	-0.01	-0.03	-0.26	-0.09	0.11	0.07	-0.04	-0.03	0.09
Baa bond	-0.05	0.00	-0.02	-0.26	-0.09	0.13	0.05	-0.04	-0.05	0.08
CP-FF spread	-0.14	0.10	0.01	-0.05	-0.13	-0.11	-0.01	-0.02	0.01	0.04

Table 1 (continued)

Variable	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10
3 mo-FF spread	-0.10	0.14	0.02	-0.10	-0.08	-0.22	-0.02	-0.05	0.00	-0.05
6 mo-FF spread	-0.13	0.12	0.02	-0.08	-0.10	-0.22	-0.02	-0.05	-0.02	-0.03
1 yr-FF spread	-0.16	0.07	-0.01	-0.07	-0.12	-0.20	-0.03	-0.06	-0.02	-0.04
5 yr-FF spread	0.04	0.07	-0.14	-0.27	-0.06	0.07	0.06	0.00	0.07	0.06
10 yr-FF spread	0.06	0.06	-0.13	-0.27	-0.02	0.06	0.05	0.01	0.08	0.06
Aaa-FF spread	0.08	0.03	-0.09	-0.25	0.04	0.05	0.02	0.01	0.05	0.11
Baa-FF spread	-0.02	0.13	0.04	-0.14	-0.02	-0.28	-0.03	-0.06	0.03	-0.07
Nominal E§ective Exchange Rate, Unit Labor Costs (IMF)	-0.01	-0.01	-0.09	-0.09	-0.06	0.18	0.04	0.11	-0.07	-0.09
Foreign Exchange Rate: Switzerland - Swiss Franc Per U.S.$	-0.02	0.04	-0.11	-0.05	-0.08	0.17	0.05	-0.02	0.06	-0.06
Foreign Exchange Rate: Japan - Yen Per U.S.$	0.00	0.03	-0.07	-0.01	-0.01	0.19	0.01	-0.01	-0.02	-0.19
Foreign Exchange Rate: United Kingdom - Cents Per Pound	0.01	-0.01	0.10	0.05	0.05	-0.14	-0.04	0.03	-0,01	0.04
Foreign Exchange Rate: Canada - Canadian $ Per U.S.$	-0.02	-0.01	-0.04	-0.09	0.02	0.08	-0.07	0.03	-0.25	0.03
Producer Price Index: Finished Goods	-0.03	-0.18	0.13	-0.07	0.04	-0.01	0.13	0.10	0.12	0.02
Producer Price Index: Finished Consumer Goods	-0.03	-0.17	0.14	-0.07	0.04	-0.01	0.15	0.10	0,11	0.04
Producer Price Index:I ntermed Mat.Supplies & Components	-0.07	-0.16	0.16	-0.05	0.02	-0.08	0.14	0.03	0.07	0.00
Producer Price Index: Crude Materials	-0.04	-0.11	0.11	0.00	0.08	-0.06	0.13	0.02	0.12	0.00
Spot market price index: bls & crb: all commodities	-0.07	0.01	0.07	-0.05	-0.04	-0.07	0.02	-0.06	0.04	-0.09
Producer Price Index: Nonferrous Materials	-0.10	0.00	0.08	-0.06	-0.04	-0.10	0.07	-0.04	0.03	-0.02
Napm Commodity Prices Index	-0.09	-0.07	0.13	0.05	-0.11	-0.10	0.08	-0.01	-0.04	-0.05
CProduction Index-U: All Items	-0.02	-0.24	0.05	-0.09	0.07	-0.01	0.11	0.07	0.08	0.04
CProduction Index-U: Apparel & Upkeep	-0.02	-0.09	-0.10	-0.03	0.03	0.01	0.00	-0.07	0.01	0.00
CProduction Index-U: Transportation	0.02	0.20	0.20	-0.01	0.02	0.02	0.15	-0.02	0.02	0.00
CProduction Index-U: Medical Care	0.01	0.22	0.20	-0.02	0.02	0.00	0.13	-0.02	0.03	0.00
CProduction Index-U: Commodities	0.02	0.21	0.20	-0.02	0.02	0.02	0.14	-0.02	0.02	0.00
CProduction Index-U: Durables	-0.03	-0.12	-0.13	0.01	-0.06	-0.05	-0.08	-0.03	-0.10	0.05
CProduction Index-U: Services	0.05	-0.17	-0.06	0.01	0.03	-0.04	0.05	0.01	0.03	-0.04
CProduction Index-U: All Items Less Food	-0.02	-0.23	0.05	-0.09	0.07	-0.03	0.11	0.05	0.09	0.05
CProduction Index-U: All Items Less Shelter	0.02	0.21	0.20	-0.02	0.02	0.01	0.14	-0.02	0.02	0.00
CProduction Index-U: All Items Less Midical Care	-0.02	-0.23	0.06	-0.09	0.07	-0.01	0.12	0.07	0.09	0.04
Pce, Impl Pr Deflator:Pce (BEA)	-0.02	-0.22	-0.04	-0.07	0.06	-0.07	0.21	0.01	0.02	-0.02
Pce, Impl Pr Deflator:Pce; Durables (BEA)	-0.02	-0.13	-0.14	0.00	-0.04	-0.11	0.02	-0.05	-0.09	0.02
Pce, Impl Pr Deflator:Pce; Nondurables (BEA)	-0.03	-0.18	0.10	-0.12	0.07	0.00	0.13	0.07	0.10	0.07
Pce, Impl Pr Deflator:Pce; Services (BEA)	0.02	-0.10	-0.13	0.03	0.03	-0.08	0.18	-0.07	-0.05	-0.12
Avg Hourly Earnings of Prod - Goods-Prod	-0.01	-0.02	0.05	-0.02	0.02	0.12	0.00	-0.08	0.03	-0.11
Avg Hourly Earnings of Prod - Construction	0.01	-0.04	0.05	0.02	-0.09	0.09	0.04	-0.05	0.06	-0.07
Avg Hourly Earnings of Prod - Manufacturi	-0.03	-0.02	0.03	-0.05	0.08	0.08	-0.04	-0.12	0.02	-0.09
U. Of Mich. Index Of Consumer Expectations (UM)	-0.01	0.10	-0.12	0.03	-0.12	0.03	-0.02	0.02	0.10	-0.12
Baseline_overall_index	-0.01	-0.06	0.12	-0.05	0.09	0.03	-0.12	0.00	-0.10	0.05
News_Based_Policy_Uncert_Index	-0.01	-0.06	0.11	-0.04	0.10	0.03	-0.11	-0.02	-0.10	0.03
Crude oil price	-0.05	-0.10	0.10	-0.07	0.01	-0.04	0.13	-0.03	0.04	0.06

In total, we estimated four different models for each commodity using backward procedures and four using forward procedures: a model that regresses futures returns on the principal components only (potentially a subset of the initial set, according to the selection procedure) in periods of low volatility and in periods of high volatility; a model that regresses futures returns on principal components and on commodity-specific factors, both in time of high and low volatility.[3] We then compute OSS forecasts for models that include and exclude commodity-specific factors using the weighted average of low-volatility and high-volatility estimates, where weights are represented by the filtered probabilities obtained from the application of the Markov switching (MS) model described earlier.

3 The data

3.1 Commodity futures returns

We consider time series of monthly returns computed using settlement prices of futures contracts on 15 commodities, collected from *Thomson Reuters Datastream,* for a period spanning from January 1989 to December 2012. The dataset contains two energy commodities (Brent Crude Oil and Gasoline), seven agricultural commodities (Corn, Soybeans,

[3] As a robustness check, instead of running two separate regressions according to a classification of the regime based on the state of the VIX, we also estimate Markov-switching predictive regressions. The results are discussed in detail in Sects. 4.3 and 5.2.

Wheat, Coffee C, Cocoa, Sugar No. 11, and Cotton No. 2), three metals commodities (Gold 100 Oz, Silver 5000 Oz, and Platinum), two general commodities (Orange Juice and Lumber), and one livestock commodity (Live Cattle).[4]

Following a common practice of both practitioners and academics, we consider investors to take fully collateralised positions in commodity futures. This implies two consequences. First, investors are not allowed to operate on margin, thus using leverage; while this approach limits the size of the return that could be reached, it has the advantage to make investments in commodities directly comparable with the investments in other asset classes, which usually require an initial money outflow. Second, the lack of a margin system limits the possibility of any unintentional liquidation (due to insufficient collateral) of the position before the end of the investor's holding period.

We compute the return on a future position on commodity j at time t as:

$$R_{j,t+1} = \frac{F^{(1)}_{j,t+1}}{F^{(1)}_{j,t}} - 1 + R^f_t, \tag{6}$$

where R^f_t is the risk-free rate between time t and $t+1$, here proxied by the 1-month T-bill rate. Naturally, the computation of the time series of futures returns is complicated by the fact that the front-end contract (typically the most liquid and used by the traders) has to be rolled over before expiry in order to maintain a long-term position (while at the same time avoiding taking a delivery). In case of physically settled contracts, to avoid the delivery, traders shall close their position before the "*First Notice Day"*, which is the first day from which the exchange may assign the delivery of the underlying asset, before the "*Last Trading Day"*. In order to identify the First Notice Day for each contract, we have considered the official trading calendar of the relevant exchange. In addition, to forecast when actually (before the First Notice Day) the majority of the investors are likely to roll over their positions, we adopted a methodology proposed by Bakshi, et al. (2019) and widely consistent with the intuition of Gorton et al. (2012) and Hong and Yogo (2012). Because the investor wants to avoid delivery, we consider that she takes position in the futures contract with the second closest maturity on the last business day of each month t, when the contract's First Notice Day occurs after the end of month t + 1.

In Table 2 we report the summary statistics of the commodity futures returns. We notice that the commodities in the sample have similar volatility, except for Corn, Cotton, Live Cattle, Gold 100 Oz, and Soybeans. Query ID="Q2" Text="Please check the layout of Table(s) 4, 5, 7, 8, 9, and correct if necessary." We also find that the time series of commodities futures returns have platykurtic tails (kurtosis lower than 3) and positive skewness, except for Corn, Live Cattle and Soybeans, which display a negative skewness. We also notice that Brent Crude Oil and Gasoline have a higher average return during the sample period than the other commodities.

3.2 Macroeconomic-based factors

Since we want to represent broad categories of economic activity, we consider the 125 macroeconomic-based factors by Ludvigson and Ng (2009), adding some series by Welch and Goyal (2008). We obtain a set of 139 variables, spanning from January 1989 to

[4] As recently shown by Aslan et al. (2018), with reference to commodity returns, it may be possible to group different commodity returns series on the basis of commonalities in the estimated linear autoregressive and non-linear threshold autoregressive features to further reduce the dimension of the cross-section.

Table 2 Summary statistics for commodity futures returns

Variable	Mean	Median	SD	Skewness	Kurtosis
Brent crude oil	0.015	0.014	0.095	0.516	3.568
Corn	−0.003	−0.007	0.074	−0.098	0.499
Lumber	−0.006	−0.008	0.095	0.575	1.684
Live Cattle	0.002	0.001	0.039	−0.481	2.720
Soybeans	0.004	0.000	0.067	−0.083	0.952
Wheat	−0.005	−0.007	0.081	0.499	2.097
Cocoa	−0.002	−0.010	0.086	0.580	1.127
Cotton No. 2	0.000	−0.004	0.076	0.375	0.863
Gold 100 Oz	0.003	−0.003	0.045	0.204	1.481
Gasoline	0.016	0.015	0.099	0.414	2.563
Orange Juice	−0.002	−0.008	0.088	0.621	1.602
Coffee C	0.001	−0.010	0.111	1.020	2.729
Platinum	0.009	0.006	0.098	0.487	3.302
Sugar No. 11	0.008	0.009	0.092	0.253	0.740
Silver 5000 Oz	0.005	−0.002	0.081	0.031	1.081

December 2012, and we group the variables into nine main categories: output and income, labour market, housing, consumption, orders and inventories, money and credit, stock market, bond and exchange rate, and prices series. A reader may find a detailed description of the variables and of the sources from which they were collected in "Appendix".

Since previous literature has highlighted that most macroeconomic series are not stationary, in the sense that they contain one (or more) unit roots, we perform the Augmented Dickey-Fuller (ADF) test on the series. The null hypothesis of this test is that the series contains a unit root; consistently, we maintain the original values of the variables if the null hypothesis is rejected at a significance level of 5%. Otherwise, we take the first difference of the variables and test them again. Again, we maintain the first difference of the variables if the null hypothesis is rejected at a significance level of 10%. By contrast, we exclude the variables if the null hypothesis cannot be rejected in the second application of the Dickey Fuller test. At the end of this process, we obtain a set of 128 macroeconomic variables.

3.3 Commodity-specific factors

The commodity-specific factors that we consider are the hedging pressure factor (HP), the basis factor, and the momentum factor, defined exactly as in Daskalaki et al. (2014). More precisely, the hedging pressure factor is represented by the difference between positive and negative hedging pressure positions. The hedging pressure of a commodity j at time t is calculated as the ratio between the number of short hedging positions minus the number of long hedging positions, divided by the total number of hedgers in the commodity market:

$$HP_{j,t} = \frac{\#shorthedgeposition_{j,t} - \#longhedgeposition_{j,t}}{Total\#hedgeposition_{j,t}}. \tag{7}$$

Table 3 Summary statistics for commodity-specific factors

Variable	Mean	Median	SD	Skewness	Kurtosis
Hedging pressure	0.004	0.005	0.044	−0.125	1.012
Basis	0.010	0.011	0.045	−0.009	1.393
Momentum	0.009	0.009	0.050	0.353	3.329

The basis factor is defined as the difference of the return of a portfolio of commodity futures with positive basis and a portfolio with negative basis. For a commodity j at time t, basis is calculated as

$$Basis_{j,t} = \frac{F_{j,t} - F_{j,t+1}}{F_{j,t}}, \tag{8}$$

where $F_{j,t}$ is the price of the nearest available futures contract on j, while $F_{j,t+1}$ is the price of the next nearest available futures contract on j.

Finally, the momentum factor is the difference between the return of a portfolio of commodity futures with highest prior 12-month return and the return of a portfolio of commodity futures with lowest prior 12-month return.[5] Gorton et al. (2012), Shen et al. (2007) and Narayan et al. (2015) have documented robust momentum effects in commodity futures returns. Table 3 reports summary statistics for the commodity-specific factors from January 1989 to December 2012.

4 Empirical results

In this Section we discuss the model estimated. Then, we provide a detailed analysis of the OOS predictability power of the models, using the MAE and the RMSFE and comparing the forecast performances of the models that include only macroeconomic variables to models based on both macroeconomic and commodity-specific factors.

4.1 Results from in-sample estimation

First, we report results of the linear regression models for the commodity futures returns. We also estimate a first order autoregressive model for each commodity futures return series, to be used as a benchmark to assess the forecast performances of the other models. In Table 4 we report the regression coefficients, the R^2 and adjusted R^2 for the models that were selected by the stepwise procedure; the regression results for low- and high-volatility states are reported separately. Low-volatility models are estimated on the period starting from January 1989 and ending in December 1997; the high-volatility models, instead, are estimated on the period from January 1998 to December 2003. The estimation periods fail to exhaust the sample because they simply form the basis for the subsequent back-testing exercise. Panels A and B of Table 4 are dedicated to backward and forward stepwise regressions that include macroeconomic variables only. Panel C and D, instead, show results for

[5] More precisely, the procedure is as follows: at any time, t, we sort the commodities according to their past 12-month performances and create an equally-weighted portfolio that is long on the first 5 commodities in the ranking and short on the last 5 commodities in the ranking.

Table 4 Estimation of backward and forward regression models

Estimation of backward and forward regression models

Panel A – Results for models including only macro principal components, backward algorithm

						LOW VOLATILITY							
Commodity Return	**Intercept**	**PC1**	**PC2**	**PC3**	**PC4**	**PC5**	**PC6**	**PC7**	**PC8**	**PC9**	**PC10**	**R-squared**	**Adjusted R-squared**
Brent Crude Oil	0.0194**							0.0133**		-0.0099	-0.0092	0.0914	0.0649
Corn	0.0006						0.0039*					0.0266	0.0173
Lumber	0.0052												
Live Cattle	0.0017		-0.0019*			0.0019						0.0490	0.0308
Soybeans	0.0025		0.0021		-0.0035*		0.0027					0.0654	0.0382
Wheat	-0.0018						-0.0041		0.0045*			0.0045	0.0323
Cocoa	-0.0098	-0.0032*										0.0300	0.0207
Cotton No.2	-0.0063	-0.0029*	-0.0029	-0.0043*		-0.0067***						0.1021	0.0668
Gold 100 Oz	-0.0033			0.0023								0.0246	0.0154
Gasoline	0.0122												
Orange Juice	-0.0030												
Coffee C	0.0060					-0.0095*				-0.0178**		0.0757	0.0579
Platinum	-0.0033	-0.0037**				-0.0062***	0.0038	-0.0074*		0.0082**		0.1863	0.1460
Sugar No.11	0.0078	-0.0035*								0.0091*		0.0562	0.0380
Silver 5000 Oz	-0.0015			0.0041				-0.0060				0.0425	0.0241
						HIGH VOLATILITY							
Commodity Return	**Intercept**	**PC1**	**PC2**	**PC3**	**PC4**	**PC5**	**PC6**	**PC7**	**PC8**	**PC9**	**PC10**	**R-squared**	**Adjusted R-squared**
Brent Crude Oil	-0.0017	-0.0086	0.0165***									0.2137	0.1905
Corn	-0.0153**					0.0043					-0.0056*	0.0736	0,0463
Lumber	-0.0375***		0.0137***		-0.0047		0.0067		0.0111*			0.2104	0.1626
Live Cattle	0.0027												
Soybeans	-0.0319*	-0.0026	0.0087**	0.0069*	-0.0062**			0.0104***	0.0060	-0.0057*		0.2082	0.1203
Wheat	-0.0511***		0.0112***	0.0096***						-0.0080***		0.2825	0.2504
Cocoa	-0.0673**		0.0192***	0.0082		0.0098*		0.0151***		-0.0065		0.2267	0.1672
Cotton No.2	-0.0159	-0.0064***	0.0077*				-0.0069*	0.0064*		-0.0054		0.2441	0.1860
Gold 100 Oz	-0.0070		0.0040*		-0.0034*							0.1275	0.1018
Gasoline	-0.0650*	-0.0096***	0.0309***	0.01306*			0.0142**	0.0092				0.3141	0.2613
Orange Juice	0.0151		-0.0082**			-0.0047*	-0.0067*	-0.0073**		-0.0103***		0.2283	0.1689
Coffee C	-0.0174						-0.0078					0.0349	0.0209
Platinum	0.0313*	-0.0067**		-0.0073	0.0061	-0.0108**					0.0143**	0.2401	0.1816
Sugar No.11	-0.0224		0.0106**	0.0061				0.0098**				0.1357	0.0970
Silver 5000 Oz	-0.0137	-0.0033*	0.0077***			0.0051*					-0.0046	0.1631	0.1124

***significant at 1% level, **significant at 5% level, *significant at 10% level

Table 4 (continued)

Estimated backward and forward stepwise regressions

Panel B – Results for models including only macro principal components, forward algorithm

Commodity Return	Intercept	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	R-squared	Adjusted R-squared
LOW VOLATILITY													
Brent Crude Oil	0.0212**							0.0139**				0.0573	0.0483
Corn	0.0006						0.0039*					0.0266	0.0173
Lumber	-0.0039			-0.0076*								0.0283	0.0191
Live Cattle	0.0017		-0.0019*			0.0019						0.0490	0.0308
Soybeans	-0.0012				-0.0032*							0.0282	0.0190
Wheat	-0.0018						-0.0041		0.0045*			0.0505	0.0323
Cocoa	-0.0098	-0.0032*										0.0300	0.0207
Cotton No.2	0.0063				-0.0033	-0.0055**						0.0840	0.0664
Gold 100 Oz	-0.0033			0.0023								0.0246	0.0154
Gasoline	0.0122												
Orange Juice	-0.0030												
Coffee C	0.0022	-0.0047				-0.0101**				-0.0158*			
Platinum	-0.0033	-0.0037**				-0.0062***	0.0038	-0.0074**		0.0082**		0.1863	0.1460
Sugar No.11	0.0078	-0.0035*								0.0091*		0.0562	0.0380
Silver 5000 Oz	-0.0015			0.0041				-0.0060				0.0425	0.0241
HIGH VOLATILITY													
Commodity Return	**Intercept**	**PC1**	**PC2**	**PC3**	**PC4**	**PC5**	**PC6**	**PC7**	**PC8**	**PC9**	**PC10**	**R-squared**	**Adjusted R-squared**
Brent Crude Oil	-0.0017	-0.0086***	0.0165***									0.2137	0.1905
Corn	-0.0153**					0.0043					-0.0056*		
Lumber	-0.0376***		0.0137***		-0.0047		0.0067		0.0111**			0.2104	0.1626
LiveCattle	0.0027												
Soybeans	-0.0011				-0.0053**			0.0041			-0.0064*	0.1403	0.1018
Wheat	-0.0021						-0.0057*	-0.0081***		-0.0091***		0.2466	0.2129
Cocoa	0.0182			-0.0072*		0.0064	-0.0134**			-0.0089*		0.1775	0.1276
Cotton No.2	0.0185*	-0.0063***		-0.0066*			-0.0112***			-0.0060		0.2233	0.1762
Gold 100 Oz	-0.0070		0.0040**		-0.0034**							0.1275	0.1018
Gasoline	-0.0009	-0.0108***	0.0181***		-0.0053					-0.0089*		0.3009	0.2585
Orange Juice	-0.0223**			0.0061**						-0.0092***		0.1685	0.1441
Coffee C	-0.0174						-0.0078					0.0349	0.0209
Platinum	0.03134*	-0.0067**		-0.0073	0.0061**	-0.0107**					0.0143	0.2401	0.1816
Sugar No.11	0.0214		-0.0083**		0.0059							0.0739	0.0466
Silver 5000 Oz	-0.0137	-0.0033***	0.0077*			0.0051*					-0.0046	0.1631	0.1124

***significant at 1% level, **significant at 5% level, *significant at 10% level

Table 4 (continued)

Estimated backward and forward stepwise regressions

Panel C – Results for models including both macro principal components and commodity-specific factors, backward algorithm

Commodity Return	Intercept	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	HP	Basis	Momentum	R-squared	Adjusted R-squared
LOW VOLATILITY																
Brent Crude Oil	0.0163							0.0116**		-0.0115*	-0.0094	-0.3885	0.1276	0.0770	0.1203	0.0675
Corn	0.0033						0.0039*					-0.0800	-0.1889	-0.0749	0.0430	0.0055
Lumber	0.0053											0.2519	0.0031	-0.0195	0.0098	-0.0190
Live Cattle	0.0005		-0.0021**			0.0019						0.0740	0.0521	0.0341	0.0620	0.0156
Soybeans	0.0037		0.0019		-0.0033*		0.0029					0.0664	-0.0585	-0.0985	0.0736	0.0180
Wheat	0.0002						-0.0039		0.0043			0.0322	-0.1588	-0.0336	0.0566	0.0099
Cocoa	-0.0120	-0.0027										0.0747	0.4010*	-0.2003	0.0665	0.0299
Cotton No.2	-0.0048	-0.0026	-0.0033	-0.0039		-0.0063***						0.1540	-0.0203	-0.0997	0.1138	0.0511
Gold 100 Oz	-0.0047			0.0026*								0.1906**	0.0965	0.0597	0.0981	0.0627
Gasoline	0.0127											-0.3510	-0.0391	0.0122	0.0287	0.0004
Orange Juice	-0.0059											0.3466	0.4232	-0.2005	0.0336	0.0054
Coffee C	0.0083					-0.0079				-0.0162*		0.5583*	-0.2175	0.0142	0.1100	0.0659
Platinum	-0.0026	-0.0036**				-0.0061**	0.0038	-0.0072*		0.0084**		0.0389*	-0.0371	-0.0180	0.1872	0.1208
Sugar No.11	0.0072	-0.0042**								0.0088*		-0.3540*	0.0011	0.0220	0.0917	0.0467
Silver 5000 Oz	-0.0014			0.004628*				-0.0050				0.3694**	-0.0252	0.1300	0.1213	0.0778
HIGH VOLATILITY																
Commodity Return	**Intercept**	**PC1**	**PC2**	**PC3**	**PC4**	**PC5**	**PC6**	**PC7**	**PC8**	**PC9**	**PC10**	**HP**	**Basis**	**Momentum**	**R-squared**	**Adjusted R-squared**
Brent Crude Oil	-0.0027	-0.0088***	0.0168***									0.1808	0.2228	-0.1531	0.2268	0.1673
Corn	-0.0158**					0.0039					-0.0053	0.2434	0.0191	0.0380	0.1135	0.0453
Lumber	-0.0354**		0.0136**		-0.0042		0.0068		0.0098*			-0.2508	0.0185	-0.1449	0.2399	0.1555
Live Cattle	0.0035											0.1182	-0.0138	-0.0220	0.0098	-0.0346
Soybeans	-0.0276	-0.0028	0.0086*	0.0067*	-0.0054**			0.0095*	0.0056	-0.0057*		0.1336	0.1057	-0.2180	0.2278	0.0991
Wheat	-0.0507***		0.012***	0.0088***						-0.0071**		0.1896	0.0858	0.0430	0.3089	0.2441
Cocoa	-0.0702**		0.0192***	0.0080		0.0105**		0.0147***		-0.0064		-0.1346	0.1119	0.0739	0.2321	0.1330
Cotton No.2	-0.0200	-0.0061***	0.0078**				-0.0072*	0.0056*		0.0051		-0.0560	0.3177	-0.0434*	0.2805	0.1876
Gold 100 Oz	-0.0039		0.0040**		-0.0027*							0.1396	0.0202	-0.1651	0.1573	0.0925
Gasoline	-0.0622*	-0.0099***	0.0305***	0.0124*			0.0138**	0.0092				0.2061	0.0424	-0.0360	0.3198	0.2321
Orange Juice	0.0214*		-0.0086***			-0.0058*	-0.0078**	-0.0078**		-0.0104***		0.3409	0.0892	-0.2811	0.2683	0.1738
Coffee C	-0.0187						-0.0075					-0.3370	-0.0280	0.0644	0.0533	-0.0041
Platinum	0.0180	-0.0062*		-0.0088*	0.0038	-0.0086					0.0161**	-0.2601	0.3922	0.3876	0.2874	0.1954
Sugar No.11	-0.0253			0.0104**	0.0055			0.0101**				-0.0313	0.0579	0.1063	0.1392	0.0585
Silver 5000 Oz	-0.0127	-0.0032*	0.0078**			0.0052*					-0.0046	-0.1397	0.0077	-0.0834	0.1828	0.0920

***significant at 1% level, **significant at 5% level, *significant at 10% level

Table 4 (continued)

Estimated backward and forward stepwise regressions

Panel D – Results for models including both macro principal components and commodity-specific factors, forward algorithm

LOW VOLATILITY																
Commodity Return	Intercept	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	HP	Basis	Momentum	R-squared	Adjusted R-squared
Brent Crude Oil	0.0199**							0.0128**				-0.3476	0.0491	0.0195	0.0801	0.0440
Corn	0.0033						0.0040*					-0.0800	-0.1889	-0.0749	0.0430	0.0055
Lumber	-0.0038			-0.0072								0.2067	0.0124	0.0082	0.0353	-0.0025
Live Cattle	0.0005		-0.0021*			0.0019						0.0740	0.0521	0.0341	0.0620	0.0156
Soybeans	7.06E-02				-2.95E+00							0.0000	0.0000	0.0000	0.0371	-0.0008
Wheat	0.0002						-0.0039		0.0043			0.0322	-0.1588	-0.0336	0.0566	0.0099
Cocoa	-0.0120	-0.0027										0.0747	0.4010*	-0.2003	0.0665	0.0299
Cotton No.2	0.0072				-0.0032	-0.0053**						0.1632	-0.0100	-0.0988	0.0979	0.0533
Gold 100 Oz	-0.0047			0.0025*								0.1906**	0.0965	0.0597	0.0981	0.0627
Gasoline	0.0127											-0.3510	-0.0391	0.0122	0.0287	0.0004
Orange Juice	-0.0059											0.3466	0.4232	-0.2005	0.0336	0.0054
Coffee C	0.0052	-0.0041				-0.0083				-0.0145*		0.4693	-0.2798	0.0741	0.1227	0.0701
Platinum	-0.0026	-0.0036**				-0.0061**	0.0038	-0.0072*		0.0084**		0.0389	-0.0371	-0.0180	0.1872	0.1208
Sugar No.11	0.0072	-0.0042**								0.0088*		-0.3540*	0.0011	0.0220	0.0917	0.0467
Silver 5000 Oz	-0.0014			0.0046*				-0.0050				0.3694**	-0.0252	0.1300	0.1213	0.0778
HIGH VOLATILITY																
Commodity Return	Intercept	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	HP	Basis	Momentum	R-squared	Adjusted R-squared
Brent Crude Oil	-0.0027	-0.0088***	0.0169***									0.1808	0.2228	-0.1531	0.2268	0.1673
Corn	-0.0159**					0.0039					-0.0053	0.2434	0.0191	0.0380	0.1135	0.0453
Lumber	-0.0354**		0.0136***		-0.0042		0.0068		0.00975*			-0.2508	0.0185	-0.1449	0.2399	0.1555
Live Cattle	0.0035											0.1182	-0.0138	-0.0220	0.0098	-0.0346
Soybeans	0.0004				-0.0047*			0.0034			-0.0057	0.0568	0.1418	-0.1623	0.1556	0.0765
Wheat	-0.0039						-0.0057*	-0.0078***		-0.0083***		0.2219	0.1537	-0.0426	0.2786	0.2110
Cocoa	0.0171			-0.0071		0.0067	-0.0137**			-0.0091		-0.0295	0.2237	-0.1418	0.1897	0.0997
Cotton No.2	0.0143	-0.0060***		-0.0068**			-0.0114***			-0.0058		-0.0301	0.3685*	-0.1013	0.2720	0.1911
Gold 100 Oz	-0.0039		0.0040**		-0.0027							0.1396	0.0202	-0.1651	0.1573	0.0925
Gasoline	-0.0004	-0.0111***	0.0183***		-0.0047**					-0.0086*		0.2104	0.1320	-0.1326	0.3086	0.2318
Orange Juice	-0.0199**			0.0058**						-0.0088**		0.2592	0.0498	-0.1216	0.1881	0.1256
Coffee C	-0.0187						-0.0075					-0.3370	-0.0280	0.0644	0.0533	-0.0041
Platinum	0.0180	-0.0062*		-0.0088*	0.0038	-0.0086					0.0161**	-0.2601	0.3922	0.3876	0.2874	0.1954
Sugar No.11	0.0177		-0.0081**		0.0055							-0.0300	0.1518	0.0396	0.0801	0.0093
Silver 5000 Oz	-0.0127	-0.0032*	0.0078**			0.005199*					-0.0046	-0.1397	0.0077	-0.0834	0.1828	0.0920

***significant at 1% level, **significant at 5% level, *significant at 10% level

Table 4 (continued)

Estimated backward and forward stepwise regressions

Panel E – AR(1) Benchmark

LOW VOLATILITY			HIGH VOLATILITY		
Commodity Return	**Intercept**	**AR(1)**	**Commodity Return**	**Intercept**	**AR(1)**
Brent Crude Oil	0.0132	0.3068***	Brent Crude Oil	0.0197	0.0523
Corn	-0.0008	0.0474	Corn	-0.0140**	-0.0707
Lumber	0.0051	-0.0122	Lumber	-0.0061	0.0466
Live Cattle	0.0041	-0.1503	Live Cattle	0.0021	0.0334
Soybeans	-0.0023	-0.0506	Soybeans	0.0038	0.1367
Wheat	-0.0006	0.0645	Wheat	-0.0123*	-0.0667
Cocoa	-0.0074	-0.0820	Cocoa	-0.0007	-0.0509
Cotton No.2	0.0064	0.0726	Cotton No.2	-0.0071	-0.1363
Gold 100 Oz	-0.0065**	-0.1012	Gold 100 Oz	0.0027	-0.0402
Gasoline	0.0120	0.3134***	Gasoline	0.0212*	-0.0009
Orange Juice	-0.0040	0.0638	Orange Juice	-0.0103	-0.1949
Coffee C	0.0074	0.1135	Coffee C	-0.0194*	-0.0924
Platinum	0.0034	0.0151	Platinum	0.0083	0.0226
Sugar No.11	0.0090	0.0697	Sugar No.11	0.0015	-0.0121
Silver 5000 Oz	-0.0035	-0.0111	Silver 5000 Oz	-0.0016	-0.3542***

***significant at 1% level, **significant at 5% level, *significant at 10% level

backward and forward algorithms that employ both macroeconomic and commodity-specific factors as predictors in a stepwise algorithm. Finally, Panel E reports the regression coefficients for the first-order autoregressive benchmark.

First, in panels A and B, backward and forward stepwise algorithms lead to the specification of rather similar models. In general, for most commodities, there is evidence of considerably more macroeconomic predictors being selected in the high-volatility regime than in the low-volatility one, consistently with our brief literature review. For instance, silver futures returns are predicted by PC3 and PC7 under both backward and forward variable inclusion algorithms in the low volatility state, and by PC1, PC2, PC5, and PC10 under both stepwise rules in the high volatility regime. The resulting adjusted R^2 coefficients are included between an approximate 2% (for gold) and 15% (platinum) in the low-volatility regime, and 2% (for coffee) and a remarkable 26% (for gasoline) in the high-VIX one.[6]

In panels C and D of Table 4, when commodity-specific predictors are added, the general insights on when and which macroeconomic factors are included as predictors remain intact. This is already an indication that the predictive power of the latter is approximately independent of the power of the former set of variables. In fact, the improvement in in-sample forecasting accuracy brought about by the basis, hedging pressure, and momentum is limited: in both panels, they are hardly ever significant in terms of t-tests, while the resulting adjusted R^2 generally declines! For instance, comparing panels A and C, we find that in the latter the adjusted R^2 coefficients now range between −2 and 12% (down from a range +2 to 15%) in the low-volatility regime, and between −4 and 20% (down from a range +2 to 26%) in the high-volatility regime. Finally, in panel E of Table 4, we report on the performance of the AR(1) benchmark, finding that the autoregressive coefficient is hardly ever significant independently of the volatility regime. Therefore, this appears to be a weak benchmark. Yet, the adjusted R^2, which we do not report because they would be hardly meaningful being based on an autoregressive structure, tended to generally exceed (although by a modest spread) those in panels A-D of the table.

4.2 Comparison of out-of-sample performances

The existing literature abounds with results in which relatively rich predictive models offer a rather accurate in-sample fit that however fails to be met by a similarly accurate OOS performance, presumably due to over-fitting problems (see, for instance, Rapach and Wohar 2006). In addition, empiricists have been aware at least since Bossaerts and Hillion's (1999) work on equity return predictability, that even predictive regressions specified on the basis of information criteria that penalize over-fitting (such as the AIC that we employ), may in any case lead to poor OOS performance. Therefore, the goal of this section is to investigate the OOS predictive power of the models estimated in Sect. 4.1, when parameter estimates are held fixed to those in Table 4 and therefore are obtained with reference to data for a 1989–2003 sample. As a result, the OOS period is Jan. 2004–Dec. 2012 and appears to be long in terms of number of observations spanned and to also include two different volatility "cycles/regimes": 2004–2007 and then 2011–2012 characterized by low volatility, and 2008–2010 characterized by a high volatility regime.

[6] The fact that the number and nature of the principal components included in the "optimized" predictive regressions is highly sensitive to whether the data are drawn from a low- versus a high-volatility regime provides indirect confirmation of the presence of regime switching dynamics in the data.

In particular, we compute the one-month ahead forecasts of futures commodity returns under both low- and high-volatility predictive regressions (specified using either backward or forward stepwise methods). This is performed for models that include macroeconomic predictors only (using the parameter estimates in panels A and B of Table 4) but also models extended to include commodity factors (using the estimates in panels C and D). At each point in time of the OOS period, the forecasts are also (i.e., in addition to forecasts that simply classify the $t+1$ regime as either low- or high-volatility, according to whether the time t filtered probability of a low-volatility regime exceeds or not 0.5) obtained as (filtered, one-step forward-iterated, real-time) probability-weighted averages of the forecasts that refer to the low- versus the high-volatility state. Under a RMSFE loss function, such weighting by predicted regime probabilities of regime-specific forecasts can be shown to be optimal under the assumption that the Markov states are independent of all other shocks in the model.

As discussed, to evaluate the accuracy of the different combinations between stepwise predictor selection techniques and whether the selection set includes or not the commodity-specific factors, we use two standard summary measures, the MAE and the RMSFE. The MAE is the average of the absolute forecast errors of the models over the OOS period, say between $t=1$ and T,

$$MAE = \frac{1}{T}\sum_{t=1}^{T}\left|\hat{y}_t - y_t\right| = \frac{1}{T}\sum_{t=1}^{T}\left|\hat{e}_t\right|, \tag{9}$$

where $\left|\hat{e}_t\right|$ is the the absolute error, $\hat{y}_t$ is the prediction from a given model/selection method, and y_t is the true value of a variable. Clearly, the smallest the MAE, the highest is the forecasting power of a model. Panel A of Table 5 reports the realized OOS values of MAE for the backward and forward stepwise methodologies. The upper portion of the table shows the values of MAE for backward and forward stepwise regressions that only include macroeconomic factors, when we do not average-weight the low- and high-volatility regime forecasts; next, we report similar, unweighted forecasts when the set of predictors also includes the basis, hedging pressure, and momentum; the last two sections of the Table report, in the same order, similar information when we weight-average the forecasts from different regimes on the basis of their one-step ahead predicted probability of being in either a low- or in a high-volatility regime.[7] The very bottom of the table reports the MAE of the AR(1) benchmark. We prefer to comment on MAE first because this forecast accuracy measure is obviously less sensitive to outliers and is therefore more robust.

In Table 5, there is no stark result, apart from the fact the *grand* averages of MAE values with and without commodity-specific factors are approximately the same. However, this just applies to the average of the MAEs (both unweighted and probability-weighted), as for some commodities there is evidence that the inclusion of the basis, hedging pressure, and momentum lowers MAE (this happens for crude oil, lumber, soybeans, gold, and platinum), while for some others it increases MAE (wheat, gasoline, orange juice, coffee, and sugar).

[7] While the upper portion of the Table relies only on whether the state probability of a low regime exceed 0.5 or not, the bottom parts of the Table also rely on the predictions from the estimated two-state MS model for the VIX. Although one may argue that this way of proceeding is more elegant and consistent with the framework of our paper, note that at this point we find ourselves jointly assessing the forecasting power of the predictive regressions that include or not commodity-specific factors and the forecasting accuracy of a simple MS model for the VIX. The latter model, as simple and compelling as it may appear, does not represent the main object of our analysis.

Table 5 Mean absolute error (MAE) and root-mean-square error (RMSFE) OOS accuracy measures

	Brent Crude Oil	Corn	Lumber	Live Cattle	Soybeans	Wheat	Cocoa	Cotton No. 2	Gold 100 Oz	Gasoline	Orange Juice	Coffee C	Platinum	Sugar No. 11	Silver 5000 Oz
Panel A: Mean absolute error															
MAE-Backward/forward selection-separated low/high volatility-commodity-specific factors excluded															
Backward-low volatility	0.0749	0.0644	0.0660	0.0336	0.0621	0.0666	0.0549	0.0613	0.0366	0.0875	0.0696	0.0725	0.0673	0.0602	0.0718
Backward-high volatility	0.1078	0.0948	0.1214	0.0318	0.1277	0.1134	0.2254	0.1632	0.0529	0.1965	0.1002	0.0743	0.1259	0.1784	0.1179
Forward-low volatility	0.0820	0.0644	0.0665	0.0336	0.0635	0.0666	0.0549	0.0638	0.0366	0.0875	0.0696	0.0762	0.0673	0.0602	0.0718
Forward-high volatility	0.1078	0.0948	0.1214	0.0318	0.1078	0.1341	0.1180	0.1212	0.0529	0.1387	0.0911	0.0743	0.1259	0.1224	0.1179
MAE-Backward/forward selection-separated low/high volatility-commodity-specific factors included															
Backward-low volatility	0.0741	0.0669	0.0665	0.0337	0.0623	0.0670	0.0600	0.0630	0.0367	0.0902	0.0719	0.0795	0.0665	0.0624	0.0705
Backward-high volatility	0.1034	0.0943	0.1131	0.0322	0.1205	0.1126	0.2224	0.1536	0.0504	0.1984	0.1057	0.0767	0.1246	0.1800	0.1188
Forward-low volatility	0.0818	0.0669	0.0656	0.0337	0.0632	0.0670	0.0600	0.0652	0.0367	0.0902	0.0719	0.0803	0.0665	0.0624	0.0705
Forward-high volatility	0.1034	0.0943	0.1131	0.0322	0.0975	0.1340	0.1173	0.1177	0.0504	0.1351	0.0923	0.0767	0.1246	0.1209	0.1188

Table 5 (continued)

	Brent Crude Oil	Corn	Lumber	Live Cattle	Soybeans	Wheat	Cocoa	Cotton No. 2	Gold 100 Oz	Gasoline	Orange Juice	Coffee C	Platinum	Sugar No. 11	Silver 5000 Oz
MAE-Backward/forward selection-low/high volatility weighted by filtered probabilities-commodity-specific factors excluded															
Backward-weighted	0.1050	0.0775	0.0893	0.0327	0.0925	0.0911	0.1345	0.1094	0.0450	0.1386	0.0852	0.0775	0.0997	0.1158	0.1189
Forward-weighted	0.1008	0.0775	0.0880	0.0327	0.0824	0.0993	0.0835	0.0912	0.0450	0.1125	0.0805	0.0791	0.0997	0.0917	0.1189
MAE-Backward/forward selection-low/high volatility weighted by filtered probabilities-commodity-specific factors included															
Backward-weighted	0.1026	0.0788	0.0859	0.0327	0.0894	0.0913	0.1348	0.1050	0.0436	0.1404	0.0902	0.0803	0.0979	0.1176	0.1182
Forward-weighted	0.0983	0.0788	0.0841	0.0327	0.0781	0.1000	0.0855	0.0891	0.0436	0.1117	0.0828	0.0805	0.0979	0.0919	0.1182
MAE-Low/high volatility weighted by filtered probabilities-AR(1) models															
AR(1)	0.0727	0.0779	0.0740	0.0325	0.0681	0.0789	0.0693	0.0736	0.0455	0.0845	0.0743	0.0714	0.0755	0.0788	0.0902
Panel B: Root-mean-square-error															
RMSE-Backward/forward selection-separated low/high volatility-commodity-specific factors excluded															
Backward-low volatility	0.0905	0.0833	0.0809	0.0414	0.0811	0.0831	0.0716	0.0812	0.0449	0.1096	0.0858	0.0961	0.0882	0.0772	0.0888
Backward-high volatility	0.1406	0.1151	0.1617	0.0395	0.1787	0.1428	0.2942	0.2243	0.0637	0.2571	0.1269	0.1033	0.1659	0.2249	0.1462
Forward-low volatility	0.1002	0.0833	0.0840	0.0414	0.0821	0.0831	0.0716	0.0829	0.0449	0.1096	0.0858	0.0975	0.0882	0.0772	0.0888
Forward-high volatility	0.1406	0.1151	0.1617	0.0395	0.1399	0.1693	0.1430	0.1604	0.0637	0.1906	0.1114	0.1033	0.1659	0.1471	0.1462

Table 5 (continued)

	Brent Crude Oil	Corn	Lumber	Live Cattle	Soybeans	Wheat	Cocoa	Cotton No. 2	Gold 100 Oz	Gasoline	Orange Juice	Coffee C	Platinum	Sugar No. 11	Silver 5000 Oz
RMSE-Backward/forward selection-separated low/high volatility-commodity-specific factors included															
Backward-low volatility	0.0871	0.0850	0.0812	0.0415	0.0815	0.0836	0.0767	0.0824	0.0451	0.1111	0.0902	0.1017	0.0873	0.0804	0.0873
Backward-high volatility	0.1357	0.1155	0.1535	0.0396	0.1688	0.1416	0.2881	0.2039	0.0614	0.2579	0.1362	0.1034	0.1630	0.2269	0.1466
Forward-low volatility	0.0976	0.0850	0.0832	0.0415	0.0824	0.0836	0.0767	0.0848	0.0451	0.1111	0.0902	0.1015	0.0873	0.0804	0.0873
Forward-high volatility	0.1357	0.1155	0.1535	0.0396	0.1246	0.1702	0.1428	0.1562	0.0614	0.1861	0.1110	0.1034	0.1630	0.1456	0.1466
RMSE-Backward/forward selection-low/high volatility weighted by filtered probabilities-commodity-specific factors excluded															
Backward-weighted	0.1386	0.0988	0.1231	0.0405	0.1370	0.1162	0.2058	0.1654	0.0553	0.1934	0.1084	0.1062	0.1344	0.1651	0.1189
Forward-Weighted	0.1294	0.0988	0.1214	0.0405	0.1120	0.1312	0.1087	0.1256	0.0553	0.1540	0.0992	0.1063	0.1344	0.1171	0.1189
RMSE-Backward/forward selection-Low/High volatility weighted by filtered probabilities-commodity-specific factors included															
Backward-weighted	0.1339	0.0994	0.1181	0.0403	0.1312	0.1160	0.2023	0.1517	0.0540	0.1938	0.1158	0.1071	0.1314	0.1669	0.1182
Forward-weighted	0.1243	0.0994	0.1163	0.0403	0.1037	0.1324	0.1101	0.1225	0.0540	0.1515	0.1015	0.1065	0.1314	0.1174	0.1182
RMSE-Low/high volatility weighted by filtered probabilities-AR(1) models															
AR(1)	0.0916	0.0965	0.0936	0.0403	0.0869	0.1019	0.0880	0.0972	0.0586	0.1116	0.0901	0.0945	0.1055	0.1022	0.1137

For the remaining five commodity futures returns series, the evidence is indeed mixed. In any event, all such differences are modest. For instance, the largest reduction occurs in the case of the Brent crude oil futures returns series, when the state probability-weighted MAE declines from 0.105 to 0.103 when commodity-specific predictors are included and a backward stepwise selection is applied (instead, in the case of the forward algorithm, the decline is from 0.101 to 0.098). However, for all series, the reported MAEs are structurally lower for the low-volatility regime, even though this regime implies stronger predictability, as one would expect from the much lower variation in realized returns of this state of the world. Finally, and crucially, for most series we observe a difficulty by predictive regressions of all types at outperforming the AR(1) benchmark, that in fact implies lower MAEs for 12 series out of 15 (the only ties are obtained for corn, live cattle, and gold, but also in these cases, the solid OOS performance is attributable to commodity factors only in the case of gold).

Panel B of Table 5 reports the OOS values for the second measure of predictive accuracy, the RMSFE, defined as

$$RMSFE = \sqrt{\frac{\sum_{t=1}^{T} \left(\hat{y}_t - y_t\right)^2}{T}}, \tag{10}$$

where the notation is the same as in Eq. (9). Panel B is then structured like Panel A. Also the key qualitative remarks expressed with reference to Panel A apply in this case. First, the *grand* average of the OOS RMSFE values with and without commodity-specific factors are approximately the same. Only in a few cases (namely, Brent crude, lumber, soybeans, especially in the high-volatility regime, and platinum), the inclusion of basis, hedging pressure, and momentum as predictors lowers the RMSFE. By contrast, for the remaining series, including commodity factors does not help. In panel B, there is some tendency for the forward stepwise algorithm to lead to lower RMSFEs than the backwards algorithm, a difference that did not appear in panel A and therefore indicates that backward-type forecasts produce larger outliers than forward-type do. Second, also in this case and for all the series, the reported RMSFEs are structurally lower for the low-volatility regime. Third, for all series but gold and live cattle futures returns, all stepwise methods and selections of predictors lead to higher RMSFEs than the AR(1) benchmark does.

4.3 Robustness checks

The analysis presented in the previous section relies on the estimation of two separate predictive regressions based on the prevailing VIX regime. One may be concerned that the use of a two-step procedure may decrease the predictive accuracy of the model because of the lower efficiency versus a one-step estimation approach. To check for this possibility, we also estimate MS predictive regressions (with and without commodity-specific factors) in which the definition of and the inferences on the state ($S_{t,j} = 1, 2$, where j indexes different commodities) are also affected by the second moment of the prediction errors, to retain a connection with second moments that so far has been captured by the VIX mean state.

The results concerning the predictive accuracy of the MS regressions are reported in Table 6. The switching predictive regressions generally lead to lower RMSFE than their probability-weighted counterparts, which is expected, as MS models are well-known to

Table 6 Robustness check—the OOS predictive accuracy of an alternative Markov switching model

	Brent Crude Oil	Corn	Lumber	Live Cattle	Soybeans	Wheat	Cocoa	Cotton No. 2	Gold 100 Oz	Gasoline	Orange Juice	Coffee C	Platinum	Sugar No. 11	Silver 5000 Oz
Markov switching model - principal components only															
MAE	0.0744	0.0795	0.0743	0.0341	0.0702	0.0857	0.0734	0.0739	0.0468	0.0898	0.0796	0.0764	0.0545	0.0827	0.0888
RMSE	0.0734	0.0812	0.0761	0.0339	0.0705	0.0850	0.0763	0.0731	0.0456	0.0913	0.0790	0.0781	0.0541	0.0830	0.0882
Markov switching model - principal components & commodity-specific factors															
MAE	0.0913	0.0977	0.0970	0.0420	0.0908	0.1108	0.0920	0.0959	0.0592	0.1168	0.0996	0.1017	0.0800	0.1054	0.1103
RMSE	0.0909	0.1008	0.0986	0.0415	0.0907	0.1125	0.0952	0.0962	0.0574	0.1192	0.1003	0.1032	0.0789	0.1048	0.1117

produce forecast errors that are less volatile than those of competing models (see, e.g., Guidolin and Timmermann 2007). Indeed, MS models that only include macroeconomic variables are often able to outperform the AR(1) benchmark, which was not the case for most of the probability-weighted predictions. By contrast, when we turn to analyze MAE, the results are qualitatively very similar to those reported in Sect. 4.2. In particular, the MS predictive regressions based on macroeconomic variables outperforms their weighted counterparts in the case of Crude Oil, Lumber, Soybeans, Wheat, Cocoa, Cotton, Gasoline, Coffee, Platinum, Sugar, and Silver, while the reverse applies to Corn, Live Cattle, Gold, and Orange Juice. However, the differences are never as stark as in the case of RMSFE. Interestingly, the adoption of MS models confirms (or even strengthens) our previous conclusion that commodity-specific factors do little to increase the predictive accuracy of the models. Instead, their inclusion always lowers both MAE and RMSFE when switching regressions are considered.

A second robustness check concerns the benchmarks used in our analysis. Indeed, as the AR(1) model turns out to be rather strong especially for some commodities such as crude oil, one may wonder whether it would be appropriate to add an AR component also to the model that includes macroeconomic predictors only, before comparing it to the model augmented with commodity-specific factors. Therefore, in a set of unreported results (that remain available upon request), we compare the RMSFE of three alternative models: (i) the simple AR(1) benchmark; (ii) a macro-PC model; (iii) a macro-PC model augmented with an AR(1) term. First, the model under (iii) hardly outperforms the macro-PC model when its OOS performance is considered: indeed, this happens only in the case of corn, cocoa, cotton, and coffee. Second, the model under (iii) never outperforms the AR(1) model. Therefore, because a model augmented with the commodity-specific factors is always able to outperform the AR(1) model, this evidence is re-assuring that it would also outperform an augmented macro model. In conclusion, this additional robustness check shows that we using a AR(1) benchmark is a conservative choice, as this is a benchmark represents a rather tall hurdle in terms of OOS performance.

5 Portfolio allocation tests

Even though the OOS statistical evidence in Sect. 4 on the predictive power of the basis, hedging pressure, and momentum as well as on all models (even those just based on macroeconomic variables) are quite grim, an investor would be more interested in the possibility to exploit the forecasts from the various models than in their MAEs or RMSFEs compared to a AR(1) benchmark. Therefore, in this Section, we conduct an additional OOS recursive asset allocation exercise based on the forecasts of the expected futures commodity returns estimated under different combinations of model selection criteria and of types of predictors included. This exercise is particularly relevant given the increasing interest of investors in including commodities in their portfolios, as they would offer diversification opportunities with respect to other asset classes with which they tend to show modest or even negative correlations (see Daskalaki and Skiadopoulos 2011; Giampietro et al. 2018). Recent literature, see e.g., Dal Pra et al. (2018) has shown a few cases in which a statistical model fails to outperform simple benchmarks in terms of realized OOS predictive accuracy and yet leads to a solid increase in risk-adjusted portfolio performance relative to the same benchmark.

5.1 Optimal portfolios in a mean–variance framework

Our aim is to compare the realized (risk-adjusted) OOS performance of portfolios that exploit forecasts of futures returns estimated with macroeconomic variables-based models with those that rely on forecasts produced expanding the set of predictors to include "local" asset class-specific variables. To use a robust framework—that over short investment intervals is known to well approximate many other types of utility modes—we compute optimal portfolios in a mean–variance set up. Among the tradable assets we include the S&P 500 total return index (as a proxy for the equity market), the US 10-Year Treasuries (to proxy the default risk-free bond market), and 30-day T-bills (to proxy cash investments). We include equity, bonds, and cash on the top of our 15 commodity futures strategies not only for realism, to simulate the strategic asset allocation decisions of a US investor over time, but also because the literature has strongly emphasized how the low correlation of commodities with the other asset classes may make them considerably more appealing than what their Sharpe ratios reveal (see, e.g., Chong and Miffre 2010; Daskalaki and Skiadopoulos 2011; Henriksen et al. 2019). To retain symmetry and obtain a fair "playing field" across assets, we apply to stock and bond returns the same predictive models (for instance, in terms of whether only macro variables are to be included, as well as in terms of the backward or forward stepwise approaches implemented) applied to commodity futures returns. For simplicity, we assume instead that the 1-month T-bill rate is a constant and known in advance.

In more detail, let $\mu_{i,t+1}$ be the predicted return with $\sigma^2_{i,t+1}$ the variance on the asset i. Suppose that an investor allocates her wealth at time t according to a set of weights $\boldsymbol{\omega}_t$, for which $\sum_{i=1}^{n} \omega_{i,t} = 1$ holds (n is the number of assets available to the investor) and that she cares (at least locally, i.e., for a one-period investment horizon) only about the conditional mean and variance of her portfolio returns, so that she wants to maximize the functional

$$\max_{\omega_t} \mu_{p,t+1} - \frac{\gamma}{2}\sigma^2_{P,t+1}, \tag{11}$$

where γ represents her risk aversion coefficient. If we indicate the risk-free rate as r^f_{t+1}, we have that, at any point in time, the portfolio expected return is

$$\mu_{p,t+1} = r^f_{t+1} + \left(\boldsymbol{\mu}_{t+1} - r^f_{t+1}{}'1\right)' \boldsymbol{\omega}_t \tag{12}$$

and the portfolio variance is

$$\sigma^2_{P,t+1} = \sum_{i,j=1}^{n} \omega_{i,t}\omega_{j,t} Cov(r_{i,t+1}, r_{j,t+1}) = \boldsymbol{\omega}_t' \boldsymbol{\Sigma}_{t+1} \boldsymbol{\omega}_t, \tag{13}$$

where $\boldsymbol{\omega}_{t+1}$ represents the vector of weights and $\boldsymbol{\Sigma}_{t+1}$ the variance–covariance matrix of asset returns predicted at time t for time $t+1$. This leads to classical, unconstrained program

$$\max_{\boldsymbol{\omega}_t} r^f_{t+1} + \left(\boldsymbol{\mu}_{t+1} - r^f_{t+1}{}'1\right)' \boldsymbol{\omega}_t - \frac{\gamma}{2}\boldsymbol{\omega}_t' \boldsymbol{\Sigma}_{t+1} \boldsymbol{\omega}_t \tag{14}$$

$$\boldsymbol{\omega}_{t+1}{}'1 = 1.$$

From its first order condition, the problem in (14) implies the formulas for the optimal vector of weights:

$$\hat{\boldsymbol{\omega}}_t = \frac{1}{\gamma} \sum_{t+1}^{-1} \left(\boldsymbol{\mu}_{t+1} - r^f_{t+1}{}'1 \right). \tag{15}$$

No short sale constraints are imposed. Following DeMiguel et al. 2007, we build static one-period optimal portfolios over time, considering expanding windows of data, starting from the beginning of our sample and including all the data until the time of the forecast; then, we calculate the corresponding portfolio expected return, realized mean–variance utility and Sharpe ratio for each time t. The recursive exercise is initialized with reference to January 1989–December 2003 to produce a January 2004 portfolio and then iterated 107 times until the last estimation sample, January 1989-November 2012, to produce an optimal portfolio for December 2012. Because in this paper we do not compute forecasts of second-order moments, we use historical data on the same expanding window described above to estimate the sample covariance matrix.[8]

For the sake of robustness, we use three different values of the risk aversion coefficient γ (0.10, 0.25, and 0.5). To make our portfolio exercise realistic, we also consider the transaction costs of rebalancing the portfolio at any time t. Additionally, we consider that an investor may want to rebalance her portfolio only if she can get an advantage in terms of the expected return of the portfolio. This means that she will decide to rebalance her portfolio at time t only if the transaction costs of rebalancing do not exceed the portfolio expected returns at $t+1$. Therefore, we solve the portfolio problem under the following, additional condition

$$\sum_{i=1}^{n} \left| \Delta\omega_{i,t} \right| \times tc_i \leq \sum_{i=1}^{n} \mu_{i,t+1} \tilde{\omega}_{i,t} \tag{16}$$

where tc is some proportional transaction cost, $\Delta\omega_{i,t}$ represents a hypothetical change in weights in case of rebalancing and $\tilde{\omega}_{i,t} \equiv \omega_{i,t-1} + \Delta\omega_{i,t}$ represents the hypothetical weights at t in case of rebalancing. If (16) does not hold, then an investor would not rebalance between t and $t+1$, as the implied costs are higher than the resulting expected benefits. Otherwise, we consider that the investor will rebalance her portfolio at t and take the resulting transaction costs into account when computing realized portfolio performance at time $t+1$. As for the imputed level of the cost tc, we face a need to introduce some simplification because an investor willing to rebalance her portfolio would pay two types of transaction costs: costs to access the market (or infrastructure costs) and liquidity costs. While it is quite difficult to make assumptions on the first type of costs, as they are likely to depend on the exact nature of an investor (e.g., whether she is an institutional investor and of what size), we can reasonably assume the liquidity costs being close to the bid-ask spread as a percentage of the mid-price and therefore being proportional to the amount transacted. Therefore, to estimate the parameter tc, we have collected daily best ask and best bid prices for commodity futures contracts, for the period Jan. 1996–Dec. 2016. For each commodity, we estimate a daily time series for tc as the bid-ask spread as a percentage of the mid-price. Next, we compute the average tc as the *grand* average of all such daily values. We get an average value of the transaction costs across all commodities equal to approximately 0.09%. Therefore, also because such estimates are considerably variable over time and across different commodities, to simplify we set $tc = 0.1\%$. When the constraint (16) is added to the general mean–variance program in (14), the solution must be performed numerically using a non-linear optimization algorithm.

[8] We have also experimented with 5-year rolling estimation windows, obtaining qualitatively similar results.

Table 7 Optimal portfolio allocations and realized performances. Panel A: The upper portion of this table reports the summary statistics for realized, optimal portfolio performances when the predictive models for commodity futures returns include only macroeconomic variables. The bottom portion contains the average weights for each asset in the asset menu. Panel B: The upper portion of this table reports the summary statistics for realized, optimal portfolio performances when the predictive models for commodity futures returns include both macroeconomic and factor-specific variables. The bottom portion contains the average weights for each asset in the asset menu. Panel C: The upper portion of the table reports summary statistics for realized, optimal portfolio performances under a benchmark AR(1) model. The bottom portion contains the average weights for each asset in the asset menu

	Backward selection $\gamma=0.1$	Forward selection $\gamma=0.1$	Backward selection $\gamma=0.25$	Forward selection $\gamma=0.25$	Backward selection $\gamma=0.5$	Forward selection $\gamma=0.5$
Panel A: Optimal portfolio allocations - macroeconomic variables models						
Average monthly return	−0.00003	0.00108	0.00000	0.00108	0.00000	0.00108
Monthly standard deviation	0.01417	0.01048	0.01401	0.01048	0.01401	0.01048
Yearly Sharpe ratio	−0.00620	0.35763	−0.00106	0.35777	−0.00091	0.35769
Mean–variance	−0.00004	0.00108	−0.00003	0.00107	−0.00005	0.00105
Average weights						
Brent crude oil	5.585%	3.434%	5.581%	3.435%	5.581%	3.435%
Corn	1.119%	2.718%	1.116%	2.718%	1.116%	2.718%
Lumber	0.578%	−0.682%	0.576%	−0.682%	0.576%	−0.682%
Live Cattle	−0.550%	−2.552%	−0.550%	−2.552%	−0.550%	−2.553%
Soybeans	1.414%	−1.235%	1.420%	−1.235%	1.420%	−1.235%
Wheat	−1.441%	−1.891%	−1.443%	−1.892%	−1.443%	−1.892%
Cocoa	−0.028%	−0.575%	−0.026%	−0.575%	−0.027%	−0.575%
Cotton No. 2	−2.749%	0.067%	−2.745%	0.067%	−2.745%	0.067%
Gold 100 Oz	3.294%	1.063%	3.289%	1.063%	3.289%	1.063%
Gasoline	−4.359%	−2.372%	−4.353%	−2.372%	−4.353%	−2.372%
Orange Juice	−0.457%	−0.453%	−0.459%	−0.453%	−0.459%	−0.453%
Coffee C	−0.020%	1.121%	−0.020%	1.121%	−0.020%	1.121%
Platinum	−0.184%	0.097%	−0.184%	0.097%	−0.184%	0.097%
Sugar No. 11	0.731%	0.604%	0.733%	0.604%	0.732%	0.604%
Silver 5000 Oz	−1.542%	−0.052%	−1.541%	−0.052%	−1.541%	−0.052%
S&P 500	2.993%	1.807%	2.992%	1.807%	2.992%	1.807%
10Y Treasury Bond	95.616%	98.902%	95.616%	98.902%	95.616%	98.902%

Table 7 (continued)

	Backward selection $\gamma=0.1$	Forward selection $\gamma=0.1$	Backward selection $\gamma=0.25$	Forward selection $\gamma=0.25$	Backward selection $\gamma=0.5$	Forward selection $\gamma=0.5$
Panel B: Optimal portfolio allocations - macroeconomic variables and commodity-specific factors models						
Average monthly return	−0.00005	0.00178	−0.00003	0.00178	−0.00003	0.00178
Monthly standard deviation	0.01554	0.00950	0.01536	0.00950	0.01535	0.00951
Yearly Sharpe ratio	−0.01053	0.64782	−0.00587	0.64782	−0.00571	0.64776
Mean–variance	−0.00006	0.00177	−0.00006	0.00177	−0.00008	0.00175
Average weights						
Brent Crude Oil	2.156%	3.855%	2.151%	3.855%	2.151%	3.855%
Corn	−0.449%	0.695%	−0.452%	0.696%	−0.452%	0.696%
Lumber	0.642%	−0.586%	0.641%	−0.586%	0.640%	−0.586%
Live Cattle	0.478%	0.474%	0.478%	0.473%	0.478%	0.473%
Soybeans	1.360%	0.171%	1.365%	0.171%	1.366%	0.171%
Wheat	−0.455%	−1.106%	−0.457%	−1.106%	−0.457%	−1.106%
Cocoa	−0.124%	−0.094%	−0.122%	−0.094%	−0.122%	−0.094%
Cotton No. 2	−1.563%	−0.206%	−1.560%	−0.206%	−1.560%	−0.206%
Gold 100 Oz	3.435%	3.006%	3.430%	3.006%	3.430%	3.006%
Gasoline	−2.030%	−3.174%	−2.024%	−3.174%	−2.023%	−3.174%
Orange Juice	0.269%	0.224%	0.268%	0.223%	0.268%	0.223%
Coffee C	0.186%	0.525%	0.186%	0.525%	0.186%	0.525%
Platinum	−0.353%	−0.461%	−0.353%	−0.461%	−0.353%	−0.461%
Sugar No. 11	−0.074%	0.296%	−0.072%	0.296%	−0.072%	0.296%
Silver 5000 Oz	−0.241%	0.221%	−0.239%	0.221%	−0.239%	0.221%
S&P 500	1.426%	1.838%	1.425%	1.838%	1.425%	1.838%
10Y Treasury Bond	95.336%	94.321%	95.336%	94.321%	95.336%	94.321%

Table 7 (continued)

	AR(1) $\gamma = 0.1$	AR(1) $\gamma = 0.25$	AR(1) $\gamma = 0.5$
Panel C: Optimal portfolio allocations - AR(1) models			
Average monthly return	0.00050	0.00050	0.00050
Monthly standard deviation	0.01332	0.01332	0.01332
Yearly Sharpe ratio	0.13080	0.13081	0.13081
Mean–variance	0.00049	0.00048	0.00046
Average weights			
Brent crude oil	3.734%	3.752%	3.752%
Corn	−1.305%	−1.307%	−1.307%
Lumber	−0.676%	−0.673%	−0.673%
Live Cattle	3.150%	3.127%	3.127%
Soybeans	0.523%	0.543%	0.543%
Wheat	−0.493%	−2.183%	−2.177%
Cocoa	−2.018%	−2.009%	−2.009%
Cotton No. 2	−0.085%	−0.098%	−0.098%
Gold 100 Oz	−26.468%	−26.381%	−26.381%
Gasoline	3.337%	3.322%	3.322%
Orange Juice	−1.000%	−1.000%	−1.000%
Coffee C	0.778%	0.772%	0.772%
Platinum	0.318%	0.329%	0.342%
Sugar No. 11	1.479%	1.469%	1.469%
Silver 5000 Oz	7.160%	7.044%	7.044%
S&P 500	2.836%	2.860%	2.860%
10Y Treasury Bond	111.869%	108.738%	108.738%

Panel A of Table 7 reports summary statistics for realized performances and recursive optimal portfolio weights obtained considering the three alternative risk aversion coefficients (0.1, 0.25, and 0.5), when the forecasts for commodity futures returns are computed from stepwise predictive regressions based on macro principal components only. Panel B reports the corresponding results when the basis, hedging pressure, and momentum are used as additional predictors. Interestingly, and almost independently of the assumed parameter γ, the largest fraction of all portfolios is allocated to 10-year Treasuries. This result seems to be plausible, given the high 10-year bond returns recorded during the 2004–2012 sample, which is in fact largely dominated by the Great Financial Crisis and the ensuing deep recession in the US, when interest rates plummeted and long-term government bonds gave extraordinarily high average returns; the demand for stocks is small but always positive. Interestingly, when commodity-specific factors are included, the share of bonds slightly declines (from 96–98% to 94–95%), the one of stocks modestly increases (from 1–2 to 2–3%), and residually the overall weight to commodities goes from 1–3% to 2–4%. In particular, crude oil, gold, and corn (when only macro factors are used) are the commodities in relatively high positive demand, and gasoline, wheat, cotton, and live cattle are the commodities in the largest negative demand.

The most striking results in Table 7, panel A, emerge from the upper portion devoted realized performances, especially when compared to panel C, that concerns portfolio weights under the benchmark model.[9] First, there is now a massive difference between the OOS realized performances of forward versus backward stepwise regression methods; in particular, while predictions based on forward stepwise regressions lead to high and appealing annualized Sharpe ratios (essentially of 0.36 independently of the selected value of γ), forecasts based on backward stepwise regressions yield essentially zero or even slightly negative risk-adjusted performances. Second, such annualized Sharpe ratios are more than double versus those obtainable under the benchmark AR(1) model (0.13 independently of γ), and this derives from both the higher realized mean returns and from lower realized portfolio standard deviations. Therefore investing in predictive technologies based on macro variables that may be thought to affect the fundamental pricing kernel does pay out in an OOS back-testing exercise, but only when the selected loss function is based on the (risk-adjusted) portfolio performance, while it does not under more classical, statistical loss functions, such as the MAE and the RMSFE covered in Table 5.

Panel B of Table 7 answers instead the question as to whether commodity-specific factors generate economic value in a portfolio problem. Although the averages for the weights for the different asset classes and commodities are generally similar (though not exactly identical) versus those in panel A, unreported plots (that remain available from the Authors upon request) reveal that their dynamics is positively correlated but not identical. As a result, the maximum realized risk-adjusted performance in the top portion of the Table differs and it is in fact considerably higher, almost double (0.65 vs. 0.36, independently of γ), than that in panel B. This implies that access to the predictive power of the basis, hedging pressure, and momentum massively increases the economic value of predictive systems applied to portfolio management that includes commodity futures among

[9] In panel C, the average allocations implied by the benchmark are even more biased towards long positions in government bonds, now exceeding 100%. The long positions in commodities are modest and now concentrated in silver, Brent crude oil, and gasoline; gold is instead massively shorted, which represents the most visible difference versus the allocations in panels A and B.

the asset classes.[10] This is consistent with a recent literature that has stressed that such "local" factors may come to play a key role in making sense of the cross-section of commodity returns, e.g., de Roon et al. (2000) and Daskalaki et al. (2014). In fact, such outperformance seems to derive more from a further improvement in the mean realized portfolio returns than from a reduction of realized risk.

Table 8 repeats the comparisons performed in Table 7, but considering the transaction costs, which in our experiment turns out to be important also because the predictive system that we propose implies a considerable degree of turnover and portfolio re-shuffling over time. Panel A is comparable to Table 7 and shows that—because an investor is allowed not to trade to rebalance her portfolio when the expected cost exceeds the expected benefit—taking realistic transaction costs into account slightly improves realized portfolio performance, in risk adjusted terms. However, this remains true only for the forward predictor selection algorithm, while for very high values of γ, some instability in the ratio appears in the case of the benchmark.[11] Moreover, our earlier conclusions concerning the economic value of the commodity-specific predictors remain intact and can be quantified in a difference between annualized Sharpe ratios of 0.65 when commodity predictors are exploited versus 0.36 when they are not, a spread of approximately 0.29.

Panels B and C of Table 8 proceed to disentangle the results under transaction costs in panel A between the periods of low- versus high-volatility.[12] Interestingly, all realized annual Sharpe ratios move up now, and the difference between the predictability regressions and the benchmark becomes massive. For instance, in periods of high volatility and for $\gamma = 1$, the benchmark achieves a ratio of 0.13 to be contrasted to a stunning 2.63 from forward stepwise regressions based on macro PCs only and 2.70 from models that also include the basis, hedging pressure, and momentum; these estimates are 0.13, 0.60, and 1.08 with reference to the low-volatility period. The finding that models of predictability generate more economic value in times of distress is fully consistent with the OOS measures of forecasting accuracy commented in Sect. 4. However, one implicit finding is that, under transaction costs, the commodity-specific factors add more risk-adjusted performance in tranquil times (the increase in Sharpe ratio is approximately 0.48 and it exceeds the probability-weighted estimate reported above) versus times of distress (the improvement is only 0.07), which implies that macroeconomic factors play a leading role especially during the less frequent crisis periods.

5.2 Robustness checks

As a robustness check, we also examine the economic value of forecasts based on the MS models introduced in Sect. 4.3. The results are reported in Table 9, where the left-portion concerns predictive regressions that exclude commodity-specific factors and the right-portion those that include them. Interestingly, the MS regressions lead to realized risk-adjusted performances that are considerably lower than those obtained from the probability-weighted model, disregarding whether commodity-specific factors are included or not.

[10] Also in this case, the effect can be noted only when the predictions are computed using a forward stepwise algorithm that starts out with a null model without any predictability, and progressively expands the set of predictors if and when these lower the AIC of the resulting model.

[11] In Table 8 and also as a way to check the robustness of our results, we have extended the exercise to include more values of the risk aversion coefficient γ, also exceeding 1.

[12] We have also performed this robustness check for the case without transaction costs and it gave insights qualitatively similar to those reported in the main text.

Table 8 Realized portfolio performances accounting for transaction costs

γ	Macroeconomic variables models						Macroeconomic variables and commodity-specific factors models						AR(1)		
	Backward selection			Forward selection			Backward selection			Forward selection					
	Yearly Sharpe ratio	Expected return	Mean–variance	Yearly Sharpe ratio	Expected return	Mean–variance	Yearly Sharpe ratio	Expected return	Mean-variance	Yearly Sharpe ratio	Expected return	Mean-variance	Yearly Sharpe ratio	Expected return	Mean–variance
Panel A: Probability-weighted strategy															
0.10	−0.00620	−0.00030	−0.00004	0.35763	0.01298	0.00108	−0.01053	−0.00057	−0.00006	0.64782	0.02133	0.00177	0.13080	0.00604	0.00049
0.25	−0.00106	−0.00005	−0.00003	0.35777	0.01299	0.00107	−0.00587	−0.00031	−0.00006	0.64782	0.02133	0.00177	0.13081	0.00604	0.00048
0.50	−0.00091	−0.00004	−0.00005	0.35769	0.01298	0.00105	−0.00571	−0.00030	−0.00008	0.64776	0.02133	0.00175	0.13081	0.00604	0.00046
0.75	−0.00121	−0.00006	−0.00008	0.35774	0.01298	0.00104	−0.00555	−0.00029	−0.00011	0.64784	0.02133	0.00174	0.13083	0.00604	0.00044
1.00	−0.00117	−0.00006	−0.00010	0.35774	0.01298	0.00103	−0.00564	−0.00030	−0.00014	0.64778	0.02133	0.00173	0.13082	0.00604	0.00041
1.50	−0.00123	−0.00006	−0.00015	0.35774	0.01298	0.00100	−0.00564	−0.00030	−0.00020	0.64780	0.02133	0.00171	1.24290	0.01724	0.00142
1.75	−0.00125	−0.00006	−0.00018	0.35779	0.01299	0.00099	−0.00567	−0.00030	−0.00003	0.64781	0.02133	0.00178	−0.08113	−0.04341	−0.02449
2.00	−0.00116	−0.00006	−0.00020	0.35780	0.01299	0.00097	−0.00568	−0.00030	−0.00003	0.64788	0.02133	0.00178	−0.08033	−0.04316	−0.02765
2.50	−0.00117	−0.00006	−0.00025	0.65520	0.01979	0.00155	−0.00571	−0.00030	−0.00003	0.53656	0.01469	0.00122	−0.07996	−0.04210	−0.03020
Panel B: Low-volatility regime															
0.10	0.82540	0.02761	0.00214	0.60086	0.02547	0.00230	1.64741	0.03211	0.00267	1.07842	0.04120	0.00343	0.13080	0.00604	0.00049
0.25	0.82545	0.33258	0.00213	0.60090	0.30853	0.00229	1.64744	0.03211	0.00267	1.07844	0.04120	0.00342	0.13081	0.00604	0.00048
0.50	0.82545	0.02571	0.00212	0.60088	0.02772	0.00227	1.64745	0.03211	0.00267	1.07843	0.04120	0.00340	0.13081	0.00604	0.00046
0.75	0.82544	0.02571	0.00211	0.60088	0.02772	0.00224	1.64747	0.03211	0.00266	1.07844	0.04120	0.00339	0.13083	0.00604	0.00044
1.00	0.82544	0.02571	0.00210	0.60086	0.02772	0.00222	1.64741	0.03211	0.00266	1.07843	0.04120	0.00337	0.13082	0.00604	0.00041
1.50	0.82556	0.02572	0.00208	0.60085	0.02772	0.00218	1.64744	0.03211	0.00265	1.07834	0.04120	0.00334	1.24290	0.01724	0.00142
1.75	0.82555	0.02572	0.00207	0.60087	0.02772	0.00215	1.64743	0.03211	0.00265	1.07844	0.04120	0.00333	−0.08113	−0.04341	−0.02449
2.00	0.82539	0.02571	0.00206	0.60086	0.30562	0.00213	1.64743	0.03211	0.00264	1.07845	0.04120	0.00331	−0.08033	−0.04316	−0.02765
2.50	0.82065	0.02549	0.00202	0.60086	0.02772	0.00209	1.64743	0.03211	0.00264	0.97875	0.02789	0.00224	−0.07996	−0.04210	−0.03020
Panel C: High-volatility regime															
0.10	1.54715	0.03249	0.00271	2.62747	0.04053	0.00338	1.38722	0.03077	0.00256	2.70347	0.03928	0.00327	0.13080	0.00604	0.00049
0.25	1.54710	0.03248	0.00270	2.62834	0.04053	0.00338	1.38585	0.03075	0.00256	2.70328	0.03928	0.00327	0.13081	0.00604	0.00048
0.50	1.55032	0.03251	0.00270	2.62876	0.04053	0.00337	1.38501	0.03074	0.00255	2.70344	0.03928	0.00327	0.13081	0.00604	0.00046

Table 8 (continued)

	Macroeconomic variables models						Macroeconomic variables and commodity-specific factors models						AR(1)		
	Backward selection			Forward selection			Backward selection			Forward selection					
γ	Yearly Sharpe ratio	Expected return	Mean–variance	Yearly Sharpe ratio	Expected return	Mean–variance	Yearly Sharpe ratio	Expected return	Mean-variance	Yearly Sharpe ratio	Expected return	Mean-variance	Yearly Sharpe ratio	Expected return	Mean–variance
0.75	1.55032	0.03250	0.00270	2.62856	0.04053	0.00337	1.38619	0.03075	0.00255	2.70350	0.03928	0.00327	0.13083	0.00604	0.00044
1.00	1.55042	0.03251	0.00269	2.62855	0.04053	0.00337	1.38623	0.03075	0.00254	2.70352	0.03928	0.00326	0.13082	0.00604	0.00041
1.50	1.55050	0.03251	0.00268	2.62859	0.04053	0.00336	1.38630	0.03075	0.00253	2.70362	0.03928	0.00326	1.24290	0.01724	0.00142
1.75	1.55048	0.03251	0.00268	2.62856	0.04053	0.00336	1.38628	0.03075	0.00253	2.70364	0.03928	0.00326	−0.08113	−0.04341	−0.02449
2.00	1.55042	0.03251	0.00267	2.62857	0.04053	0.00336	1.38599	0.03075	0.00252	2.70367	0.03928	0.00326	−0.08033	−0.04316	−0.02765
2.50	1.55044	0.03251	0.00266	2.62847	0.04053	0.00335	1.38628	0.03075	0.00251	2.70396	0.03929	0.00325	−0.07996	−0.04210	−0.03020

Table 9 Robustness check—optimal portfolio allocations under an alternative Markov switching model. The upper portion of the table reports the summary statistics for realized, optimal portfolio performances when the predictive models for commodity futures returns include only macroeconomic variables (on the left side) and when it includes both macroeconomic and factor-specific variables (on the right side). The bottom portion contains the average weights for each asset in the asset menu

	Principal components only			Principal components & commodity-specific factors		
	$\gamma=0.1$	$\gamma=0.25$	$\gamma=0.5$	$\gamma=0.1$	$\gamma=0.25$	$\gamma=0.5$
Average monthly return	0.0014	−0.0004	0.0087	−0.0062	−0.0064	−0.0119
Monthly standard deviation	0.0437	0.0423	0.0552	0.0720	0.0713	0.0512
Yearly Sharpe ratio	0.1100	−0.0306	0.5429	−0.2971	−0.3132	−0.8032
Mean–Variance	0.0013	−0.0006	0.0079	−0.0064	−0.0071	−0.0125
Average weights						
Brent crude oil	4.638%	3.384%	1.904%	2.854%	−1.732%	3.900%
Corn	1.490%	1.851%	5.879%	5.430%	4.287%	8.732%
Lumber	0.576%	0.699%	0.088%	2.640%	0.895%	1.257%
Gasoline	−0.678%	0.245%	3.406%	3.735%	10.098%	7.046%
Soybeans	−1.623%	−2.344%	−4.384%	−1.254%	0.199%	−6.914%
Wheat	1.500%	1.931%	−0.635%	1.918%	−0.111%	1.242%
Cocoa	−1.183%	−1.345%	−0.512%	−1.304%	−2.624%	−3.373%
Cotton No. 2	−2.686%	−1.663%	−2.254%	−2.912%	−0.018%	0.229%
Gold 100 Oz	1.424%	3.654%	−3.241%	2.446%	2.911%	2.194%
Gasoline	−0.678%	0.245%	3.406%	2.423%	6.503%	−1.450%
Orange Juice	0.238%	−0.005%	−0.040%	−1.232%	0.525%	0.238%
Coffee C	2.093%	2.304%	1.992%	0.442%	3.365%	2.527%
Platinum	−3.946%	−5.398%	−5.757%	−3.779%	−10.129%	−3.402%
Sugar No. 11	1.391%	1.825%	2.516%	2.576%	2.241%	0.843%
Silver 5000 Oz	4.353%	3.483%	6.526%	4.850%	6.434%	6.236%
S&P500	12.820%	10.749%	13.653%	14.471%	4.338%	5.012%
10Y Treasury Bond	72.047%	73.515%	73.901%	66.695%	72.817%	75.683%

Indeed, the highest Sharpe ratio that we obtain from the switching predictive regressions is 0.11 (for an investor with a modest risk aversion coefficient of 0.1 and when only macroeconomic variables are included). This is one sixth of the best Sharpe ratio that we achieve under the weighted-regressions, which equals 0.65 (for all the investors and when the commodity-specific factors are included). This seems to confirm our intuition that models that outperform when statistical loss functions are applied may fail to deliver superior risk-adjusted performances when used to produce optimal portfolio allocations.[13]

[13] We also performed the exercises accounting for transaction costs, similarly to Sect. 5.1. The results, which are not reported for the sake of brevity, are comparable to those discussed for the case of no transaction costs.

6 Conclusions

In this paper, we have compared the predictive power for commodity futures returns of models based on macroeconomic variables with models augmented to include three commodity-specific factors that have received considerable attention in earlier work. Differently from previous papers, we concentrate almost entirely on the genuine OOS forecasting power of a range of regressions (compared to a simple AR benchmark). Finally, we have assessed the relative performance of the models when they are used to produce forecasts to be employed in a mean–variance framework.

In particular, the primary objective of this paper was to assess whether commodity-specific predictors are needed to accurately forecast commodity returns (thus supporting the idea that commodities are an “alternative”, per-se asset class) or instead macroeconomic variables are sufficient (and hence commodities are similar to the rest of the asset classes). Our analysis leads us to conclude that the relevance of commodity-specific factors depends on whether we use statistical or economic loss functions. Indeed, while commodity-specific factors do not seem to be relevant when OOS predictive accuracy measures are considered, they become of paramount importance when we use the predictions to build recursive, static, one-period mean-variance portfolios. More specifically, from the in-sample estimation exercises, we conclude that neither models based on macroeconomic variables alone, nor models which also include commodity-specific factors outperform the others. Both types of predictive regressions lead to small adjusted R^2; furthermore, all models imply the widespread appearance of non-significant coefficients for most commodity-specific factors as well as a majority of the macro principal components. Probably as a result of this, when we investigate the OOS predictive accuracy of the models over the 2004–2012 period, we find that the MAE and the RMSFE scores fail to show any marked differences across models. In fact, all types of models perform worse than the benchmarks, with exception of the predictions generated in the low-volatility regime.

Against this background, the OOS results from the recursive portfolio allocations and the resulting risk-adjusted performances are instead sharp: exploiting predictive regressions—remarkably those from stepwise forward algorithms that go “simple to general”—tends to generate large increases in realized Sharpe ratios. Consistently with the idea that commodities represent a peculiar asset class, the ability to create economic value is further enhanced when the basis, hedging pressure, and momentum are used as predictors, especially in times of quiet financial markets and low volatility. The results are even starker when transaction costs are taken into account (and the investor has the possibility to choose to trade only when the expected benefit exceed the rebalancing cost).

Even though they can generate economic value (and this is a notable finding that should be of interest for money and risk managers), it remains the case that in statistical terms, both the macroeconomic and commodity-specific factors carry limited predictive power for commodity futures returns. A potential explanation for this empirical finding is that commodities are extremely heterogeneous in terms of their economic determinants, and therefore a unique, even if detailed, set of variables may be insufficient to capture and predict futures contract returns. In future research, it may prove useful to investigate whether the inclusion of commodity factors specific *to each underlying*, individual commodity (e.g., oil inventory for oil futures, temperature levels in specific areas for orange juice output, rainfall levels in specific regions for soybean crops, etc.) may deliver further, substantial increases in forecasting power.

In this paper, we have adopted a pseudo-out of sample approach in that the predictor selection methodology implemented through stepwise regressions is only performed in the full sample and again with reference to the first date of our recursive back-testing sample (December 2003). It would be interesting to adopt online algorithms stemming from the theory of attribute distributed learning (a particular class of machine learning algorithms, see Zheng et al. 2013) that sequentially takes new observations and incorporates them immediately, simultaneously adjusting the way that the individual predictors are combined and providing feedback to the individual predictors for them to be retrained in order to achieve a better ensemble predictor in real time. Zheng et al. (2013) have proven that such algorithms are particularly useful when applied to financial market prediction.

Finally, although our results show that there may exist a payoff from modelling commodity futures returns (selecting the corresponding linear regressions by stepwise methods), it would seem natural to explore whether other aggregate variables (possibly also related to commodity markets) may improve the overall forecasting performance. Our set merging Ludvigson and Ng's (2009) with Welch and Goyal (2008) variables appears to be rich, but richer is always possible. Moreover, while in this paper we have classified sample dates as belonging to good and bad times and then, conditioning on that, distinguished between regimes of low- and high-volatility, it may prove interesting to develop an integrated regime switching predictive framework in which regimes and forecasting models are specified and estimated jointly, as in Giampietro et al. (2018).

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Appendix

The first group of macroeconomic variables includes output and income series, such as personal income, retail sales, and real consumption, taken from The Conference Board's Indicator (TCB), and total industrial production, extrapolated from the Global Insights Basic Economics Database (GIBED). The second group contains labour market series. The series of average weekly hours of production or non-support workers on private non-farm payrolls, total employees in civilian labour force, and unemployment average duration in weeks are taken from the database GIBED, while the source of average weekly initial claims for unemployment insurance is TCB. The third group contains housing series. The series of total new private housing units are taken from GIBED, while the manufacturing and trade inventories are taken from TCB. The fourth group includes consumption, orders, and inventories series, such as manufacturers' new orders series and National Association Of Purchasing Managers (NAPM) new orders index, National Association Of Purchasing Managers (NAPM) vendor deliveries index, and National Association Of Purchasing Managers (NAPM) inventories index. The fifth group includes money and credit variables. The series of monetary base, S&P 500, commercial and industrial loans, and average effective exchange rates of this group are taken from GIBED. To this series, we add the historical news-based policy index and the economic policy uncertainty index. The historical news-based policy index is a proxy of the economic policy uncertainty for the US, based on three types of underlying components: the newspaper coverage of policy-related economic uncertainty, the number of federal tax code provisions set to expire in future years, and the disagreement among economic forecasters. The economic policy uncertainty index represents the US movements in policy-related economic uncertainty. This index indicates that when the uncertainty increases, stock prices rise and the level of investment, employment, and country's outcome go down.

The sixth group includes stock market series. They are S&P 500 Index and S&P 500 industrials series. We also include dividend yield (DY), earning price ratio (EP), and dividend pay-out ratio (DP) of S&P 500 index. The dividends used here are the 12-month moving sums of dividends paid on the S&P 500 Index. Earning price ratio is obtained as the log of earnings minus the log of stock prices where the earnings are the 12-month moving sums on the S&P 500 Index. The dividend pay-out ratio is computed as the log of dividends minus the log of earnings. We also consider stock return volatility (SVOL), which is the monthly sum of the squared daily stock returns on the S&P 500 index, and book-to-market ratio (BM), that is the ratio of the book value to market value for the Dow Jones Industrial Average. We also take into account the net equity expansion (NTIS), which is the ratio of the 12-month moving sums of net issues by New York Stock Exchange (NYSE) listed stocks to the total market capitalization of NYSE stocks.

The seventh group includes bond and exchange rate series. We additionally consider Treasury Bill Rate (TBL), which relates to the interest rate on a three-month Treasury bill, the long-term government bond yield (LTY), the long-term return government bond (LTR) returns, the term spread (TMS), the default yield spread (DFY), calculated as the difference between Moody's BAA- and AAA-rated bond yields, and the default return spread (DFR), calculated as the difference between long-term corporate and government bond returns.

The eighth group includes prices series. They represent producer price index series, National Association of Purchasing Managers (NAPM) commodity price index, consumer of price index series, personal consumption expenditures series, and average hourly earnings of production and nonsupervisory employees series. To these, we add the crude oil price (COP), computed using the logarithmic changes in the nominal price of West Texas Intermediate crude oil, provided by the Federal Reserve Bank of St. Louis.

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